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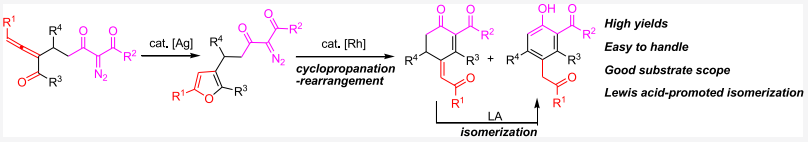
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Synthesis and Rhodium(II)-Mediated Cascade Cyclopropanation/Rearrangement/Isomerization of Diazo 2,3,5-Trisubstituted Furans: The Construction of Penta-substituted Aromatic Compounds

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Cite This: J. Org. Chem. 2020, 85, 2395−2405 Read Online

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ABSTRACT: Ag(I)-catalyzed synthesis of diazo-trisubstituted furans starting from diazo-cumulated allenyl ketones has been investigated. The $Rh_2(OAc)_4$-catalyzed reaction of the diazo 2,3,5-trisubstituted furans provided penta-substituted aromatics via cascade intermolecular cyclopropanation/rearrangement/isomerization. The cyclopropanation on the furan ring/rearrangement of cyclopropane moiety has been reported. A reasonable mechanism is proposed.

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1. INTRODUCTION

A diazo compound is a sort of important intermediates and have been extensively employed as a carbene source in organic synthesis. Many achievements have been made in the synthesis and application of diazo compounds.[1−5] For example, the functionalization of nonactivated C−H bonds,[6] cyclopropanation reaction[7−9] (Scheme 1a), and C(sp^2)−H insertion reaction[10−15] (Scheme 1b) has been described in literatures. However, the cyclopropanation on the furan ring via metal carbenoid has rarely been reported.[16] In fact, to the best of our knowledge, synthetic application via the cyclopropanation has never been reported.

In addition, penta-substituted aromatic compounds are important intermediates in organic synthesis.[17] Synthesis of the aromatic compounds has been widely explored, and many achievements have been made.[18] However, they generally suffer from some drawbacks such as multistep reactions.[19] Therefore, the investigation on their simple and efficient synthesis is always underway.

Based on these considerations, a novel protocol for the construction of penta-substituted aromatic compounds via a selective cyclopropanation on the furan ring without the corresponding C(sp^2)−H insertion is of interest. Herein, we reported the synthesis of diazo-trisubstituted furans and their application in the construction of penta-substituted aromatic compounds (Scheme 1c).

2. RESULTS AND DISCUSSION

Diazo-cumulated allenyl ketones **1aa**−**1gb** were easily prepared by Michael reaction of α-alkynyl α,β-unsaturated ketones with β-keto α-diazo carbonyl compounds based on the method reported in literature.[20] The synthesis of diazo-trisubstituted furans became our first objective. For this goal, ethyl 6-benzoyl-2-diazo-3-oxo-5,8-diphenylocta-6,7-dienoate **1aa** was chosen as the model substrate to establish a reaction system for the construction of the furan cycle.[21]

On the base of our previous research,[15] an initial plan for the construction was to use Ag(I) compounds as the catalyst. In the AgI (or $AgSbF_6$)/CH_3CN/90 °C system, these attempts failed (Table 1, entries 1−2). Happily, the utilization of $AgNO_3$ offered the desired product **2aa**, albeit in low yield (Table 1, entry 3). Other silver(I) compounds such as AgOAc, CF_3CO_2Ag, and $AgBF_4$ could also achieve the transformation (Table 1, entries 3−6). In addition, the Pd(0) catalyst was exploited as well. In the $Pd(PPh_3)_4$/THF/N_2/25 °C system, the product **2aa** was indeed generated, but with accompany of the formation of an unknown by-product (Table 1, entry 7). Among all catalyst tested, CF_3CO_2Ag afforded the best result in terms of yield and cleanness of the reaction (Table 1, entry 6).

In the CF_3CO_2Ag (20 mol %)/25 °C system, the solvent effect on this reaction was examined to increase the yield (Table 1, entries 8−12). The investigation of solvent revealed that the highest yields (88%) were achieved in DCE (Table 1, entry 9). Reaction temperature also affected the reaction progress. As expected, the elevation of reaction temperature facilitated to shorten reaction time (Table 1, entries 13−15). However, the yield decreased at 80 °C, owing to the formation of unknown side product (Table 1, entry 15).

On the other hand, the effect of catalyst loading of CF_3CO_2Ag between 5 and 20 mol % was exploited (Table 1, entries 14, 16−

Received: November 15, 2019
Published: January 9, 2020

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https://dx.doi.org/10.1021/acs.joc.9b03093
J. Org. Chem. 2020, 85, 2395−2405

Scheme 1. Reaction of Diazo Carbonyl Compounds and C(sp^2)−H Carbenoid Insertion

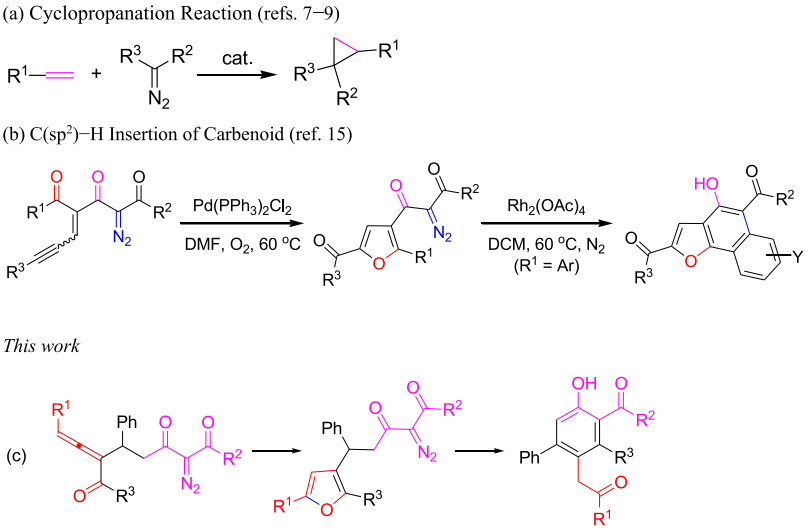
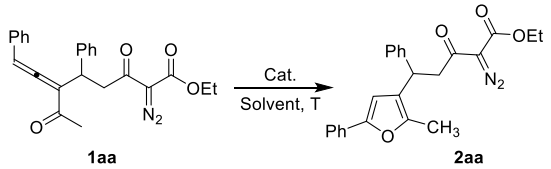


Table 1. Reaction Condition Optimization for the Synthesis of Diazo Furan Derivatives[a]



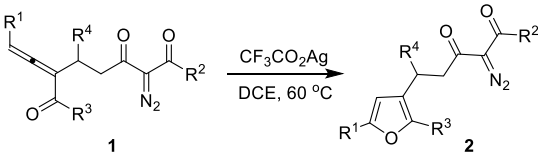
entry	cat.	solvent	T (°C)	t (h)	yield (%)[b]
1	AgI (10 mol %)	CH_3CN	90	18	NR[c]
2	$AgSbF_6$ (10 mol %)	CH_3CN	90	20	NR[c]
3	$AgNO_3$ (10 mol %)	CH_3CN	90	7	40[d,e]
4	AgOAc (10 mol %)	CH_3CN	90	15	30[e]
5	$AgBF_4$ (10 mol %)	CH_3CN	90	20	40[e]
6	CF_3CO_2Ag (20 mol %)	CH_3CN	90	16	42[e]
7	$Pd(PPh_3)_4$ (10 mol %)	THF	25	66	30[e,f,g]
8	CF_3CO_2Ag (20 mol %)	DCM	25	8.5	86[h]
9	CF_3CO_2Ag (20 mol %)	DCE	25	8.5	88[h]
10	CF_3CO_2Ag (20 mol %)	THF	25	35	24[e,g]
11	CF_3CO_2Ag (20 mol %)	toluene	25	31	58[e]
12	CF_3CO_2Ag (20 mol %)	1,4-dioxane	25	30	61[e]
13	CF_3CO_2Ag (20 mol %)	DCE	40	2.5	80[h]
14	CF_3CO_2Ag (20 mol %)	DCE	60	0.5	82[h,i]
15	CF_3CO_2Ag (20 mol %)	DCE	80	0.4	80[h,i]
16	CF_3CO_2Ag (5 mol %)	DCE	60	5	75[h]
17	CF_3CO_2Ag (10 mol %)	DCE	60	2	80[h]
18	CF_3CO_2Ag (15 mol %)	DCE	60	1.5	80[h]

[a]Reaction conditions: ethyl 6-acetyl-2-diazo-3-oxo-5,8-diphenylocta-6,7-dienoate **1aa** (0.2 mmol, 80 mg), catalyst in various of solvent (3 mL) at different temperature. [b]Isolated yields by column chromatography. [c]No reaction. [d]No light. [e]The substrate was not fully converted. [f]Under nitrogen atmosphere. [g]The formation of by-product. [h]Substrate **1aa** was fully consumed. [i]The product **2aa** may be transformed into unknown side products in the prolonged reaction time.

18). With the increase of catalytic loading, faster and cleaner reaction occurred. The best result was obtained at 20 mol % loading based on both yield and reaction time (Table 1, entry 14). Thus, the optimal conditions for the formation of **2aa** were obtained as follows: the use of CF_3CO_2Ag (20 mol %) as the catalyst and DCE as the solvent at 60 °C.

With the optimized reaction conditions in hand, we next investigated the substrate scope for the formation of **2**. These results are summarized in Table 2. In the case of R^2 = OEt, the influence of the R^1 group, which is adjacent to the allene moiety, on the reaction was explored first. For aromatic, heterocyclic, alkyl group, and hydrogen (R^1), diazo-trisubstituted furans **2aa**−**2ja** were generated in moderate to excellent yields of 65−95% (Table 2, entries 1−10). In the case of R^2 = Ph, the treatment of substrates **1ab**−**1gb** with CF_3CO_2Ag also gave the

Table 2. Substrate Scope for the Formation of Diazo-Trisubstituted Furans 2[a]



entry	product	R^1	R^2	R^3	R^4	t (h)	yield (%)[b]
1	**2aa**	Ph	OEt	Me	Ph	0.5	82
2	**2ba**	p-MeC_6H_4	OEt	Me	Ph	0.4	78
3	**2ca**	p-$MeOC_6H_4$	OEt	Me	Ph	0.5	80
4	**2da**	p-BrC_6H_4	OEt	Me	Ph	0.6	81
5	**2ea**	p-ClC_6H_4	OEt	Me	Ph	0.4	80
6	**2fa**	p-FC_6H4	OEt	Me	Ph	0.5	78
7	**2ga**	3-thienyl	OEt	Me	Ph	0.5	80
8	**2ha**	n-Bu	OEt	Me	Ph	0.4	95
9	**2ia**	t-Bu	OEt	Me	Ph	0.3	80
10	**2ja**	H	OEt	Me	Ph	1	65
11	**2ab**	Ph	Ph	Me	Ph	0.5	80
12	**2bb**	p-MeC_6H_4	Ph	Me	Ph	0.5	80
13	**2cb**	p-FC_6H_4	Ph	Me	Ph	0.5	75
14	**2db**	3-thienyl	Ph	Me	Ph	0.5	79
15	**2eb**	n-Bu	Ph	Me	Ph	0.4	82
16	**2fb**	t-Bu	Ph	Me	Ph	0.6	85
17	**2gb**	H	Ph	Me	Ph	0.5	79

[a]Reaction conditions: substrates **1** (0.2 mmol) and CF_3CO_2Ag (20 mol %) in DCE (3 mL) at 60 °C. [b]Isolated yields by column chromatography.

https://dx.doi.org/10.1021/acs.joc.9b03093
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corresponding products **2ab**−**2gb** in yields of 75−85% (Table 2, entries 11−17). In addition, we tried to expand the scope of the reaction using designed substrates **1** (R^3 = H, other alkyl or aryl; R^4 = H, alkyl or substituted phenyl), but the attempt was unsuccessful because the preparation of these substrates has not been reported in literature and has also been not achieved by us.

Next, we began the exploration of subsequent cycloaddition process of diazo-trisubstituted furans **2**. In theory, $Rh_2(OAc)_4$-mediated reaction of compound **2** could undergo Wolff rearrangement and C(sp^2)−H insertion,[15] providing the corresponding products **P-I**, **P-II** and **P-III** (Figure 1). On the

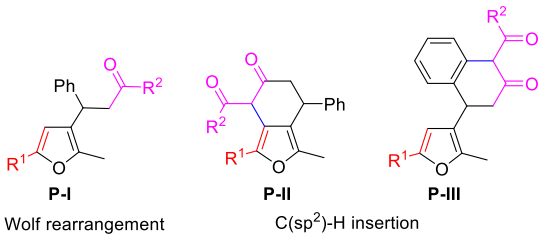


Figure 1. Possible Products from Diazo-Trisubstituted Furans **2**.

basis of the protocol, we explored the possible cyclization reaction of diazo-trisubstituted furans **2**. Unfortunately, the corresponding Wolff rearrangement and C(sp^2)−H insertion products were not detected for all substrates examined by us. Interestingly, in most cases, the reaction offered penta-substituted aromatic compounds **4** or nonisomerization products **3**. These results arre illustrated in Table 3.

For substrates **2aa**, **2fa**, and **2ha**−**2ja** (R^1 = Ph, *p*-FC_6H_4, *n*-Bu, *t*-Bu, and H), the reaction gave the corresponding nonisomerization products **3aa**, **3fa**, and **3ha**−**3ja** (Table 3, entries 1, 6, and 8−10). In the case of R^2 = OEt, whether R^1 is the aryl group (*p*-MeC_4H_4, *p*-$MeOC_4H_4$, *p*-BrC_4H_4, *p*-ClC_4H_4) or 3-thienyl group, the only isomerization products **4ba**−**ea** and **4ga** were obtained without the corresponding nonisomerization products **3ba**−**ea** and **3ga** (Table 3, entries 2−5 and 7). We speculate that the aromatization of compounds **3ba**−**ea** and **3ga** occurred more easily than that of compounds **3aa**, **3fa**, and **3ha**−**3ja**. However, the reason is not clear at present.

When R^2 was changed into the phenyl group, the reaction only afforded nonisomerization products **3ab**−**3cb**, **3eb**, and **3gb** (Table 3, entries 11−13, 15, and 17). For substrates **2db** and **2fb** (R^1 = 3-thienyl and *t*-Bu), no desired products **3** and **4** were obtained, perhaps owing to both the steric effect of *t*-butyl group and the electron effect of Ph group or 3-thienyl (R^2). In the two cases, a mixture of complicated unknown side products was only generated (Table 3, entries 14 and 16). In general, when R^2 is Ar, the compound **3** is relatively stable because of the existence of great π-conjugated system; in the case of R^2 = OEt, the stability of product **3** becomes weak. Thus, the aromatization of the latter into the corresponding product **4** is easier to carry out than that of the former. The structure of the penta-substituted aromatic compound was finally established by single-crystal X-ray diffraction analysis of product **4ea** (CCDC 1876455).

An attempt was made to transform compounds **3** into penta-substituted aromatic compounds **4**. An important influence of R^1 and R^2 on the transformation was detected in the $BF_3{\cdot}OEt_2$/DCM/25 °C system. These results are summarized in Scheme 2.

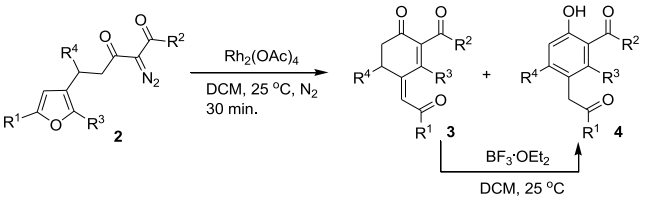
In the case of R^2 = OEt, the effect of the substituent R^1 on the reaction was explored. The treatment of the substrate (**3aa**, R^1 = Ph; **3fa**, R^1 = *p*-FC_6H_4; **3ha**, R^1 = *n*-Bu; **3ia**, R^1 = *t*-Bu) gave the corresponding penta-substituted aromatic compounds **4** (Scheme 2, **4aa**, **4fa**, **4ha**, and **4ia**). In the case of R^2 = Ph, *p*-FC_6H_4, and *n*-Bu, the conversion has been also achieved to offer the corresponding products (Scheme 2, **4ab**, **4cb**, and **4eb**). Obviously, the isomerization of compound (R^2 = Ph) **3** into **4** was more difficult than that in the case of R^2 = OEt, perhaps owing to their exceptional difference in stability.

Regretfully, when R^2 is hydrogen, only a complicated mixture of side products was formed under the standard conditions (Scheme 2, **4ja** and **4gb**). In the case of R^1 = *p*-MeC_6H_4/R^2 = Ph, a similar result was observed (Scheme 2, **4bb**). However, the mechanism for the influence of R^1 and R^2 on the isomerization reaction is not clear at present.

In addition, a possibility for one-pot synthesis of penta-substituted aromatic compounds **4** starting from ethyl 6-acetyl-2-diazo-3-oxo-5,8-diphenylocta-6,7-dienoate **1aa** has been exploited. In the $Rh_2(OAc)_4/CF_3CO_2Ag/BF_3{\cdot}OEt_2$/DCM/$N_2$/25 °C system, that is, under the optimal conditions for above-mentioned step-by-step reaction, the one-pot reaction only gave the corresponding furan derivative **2aa** in 50% yield. The conversion of **2aa** into **3aa** and **4aa** did not occur, perhaps because of the deactivation of $Rh_2(OAc)_4$ in the system (Scheme 3).

Based on our results and reports in literatures,[22] a plausible mechanism for the formation of cyclization products **4** starting from trisubstituted furans **2** is depicted in Scheme 4. First, the $Rh_2(OAc)_4$-promoted decomposition of diazo compounds **2** affords Rh(II)-carbene **IA**. The Rh(II)-carbene **IA** undergoes an

Table 3. Scope of $Rh_2(OAc)_4$-Mediated Cyclization of Diazo-Trisubstituted Furans[a]

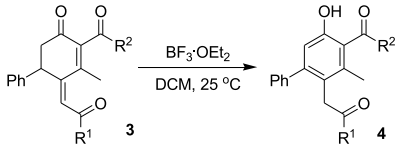


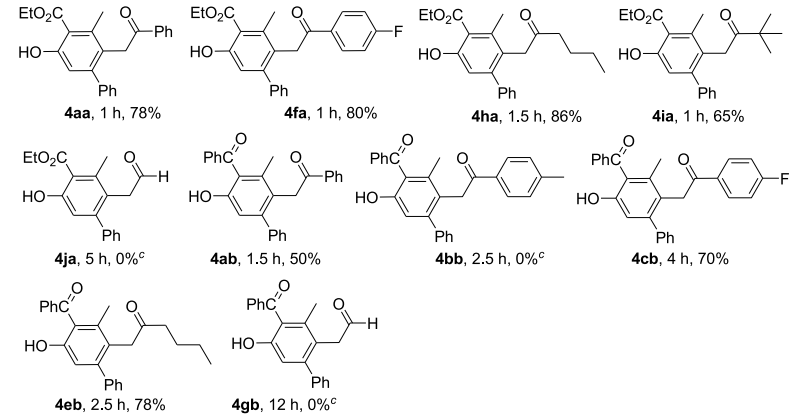
entry	product	R^1	R^2	R^3	R^4	ratio (3/4)	yield (%)[b]
1	**aa**	Ph	OEt	Me	Ph	100:0	77
2	**ba**	*p*-MeC_6H_4	OEt	Me	Ph	0:100	68
3	**ca**	*p*-$MeOC_6H_4$	OEt	Me	Ph	0:100	82
4	**da**	*p*-BrC_6H_4	OEt	Me	Ph	0:100	68
5	**ea**	*p*-ClC_6H_4	OEt	Me	Ph	0:100	73
6	**fa**	*p*-FC_6H_4	OEt	Me	Ph	100:0	82
7	**ga**	3-thienyl	OEt	Me	Ph	0:100	80
8	**ha**	*n*-Bu	OEt	Me	Ph	100:0	84
9	**ia**	*t*-Bu	OEt	Me	Ph	100:0	60
10	**ja**	H	OEt	Me	Ph	100:0	82
11	**ab**	Ph	Ph	Me	Ph	100:0	42
12	**bb**	*p*-MeC_6H_4	Ph	Me	Ph	100:0	65
13	**cb**	*p*-FC_6H_4	Ph	Me	Ph	100:0	70
14	**db**	3-thienyl	Ph	Me	Ph		0[c]
15	**eb**	*n*-Bu	Ph	Me	Ph	100:0	75
16	**fb**	*t*-Bu	Ph	Me	Ph		0[c]
17	**gb**	H	Ph	Me	Ph	100:0	40

[a]Reaction conditions: furan derivatives **5** (0.2 mmol) and $Rh_2(OAc)_4$ (10 mol %) in DCM (3 mL) at 25 °C under a nitrogen atmosphere. [b]Isolated yields by column chromatography. [c]By-product was produced.

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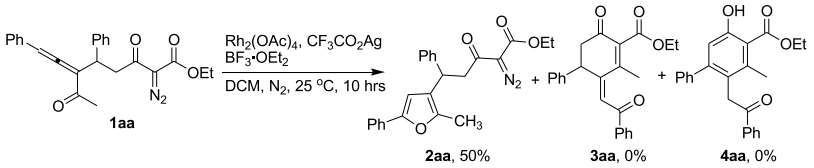
Scheme 2. Scope for Isomerization of Compounds 3 into Penta-substituted Aromatic Compounds 4[a,b]



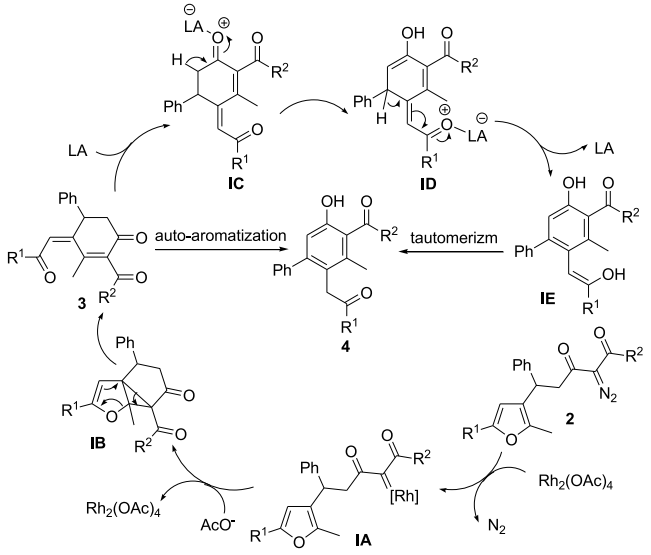


[a]Reaction conditions: **3** (0.2 mmol) and BF_3·OEt_2 (1.0 equiv) in DCM (2 mL) at 25 °C for several hours. [b]Isolated yields by column chromatography. [c]By-product was formed.

Scheme 3. Attempt for One-Pot Synthesis of Penta-substituted Aromatic Compound 4



Scheme 4. Reasonable Mechanism for the Construction of Penta-substituted Aromatic Compound 4



intramolecular cyclopropanation on the furan ring to form intermediate **IB**.[16] Then, a rearrangement reaction of the intermediate **IB** occurs via cyclopropane ring-opening reaction, leading to the formation of six-membered ring compound **3**. An attempt for the isolation of the intramolecular cyclopropanation intermediate **IB** was unsuccessful because of their great unstability. The auto-isomerization of relatively unstable compound **3** offers penta-substituted aromatic compounds **4**. For relatively stable compound **3**, BF_3-promoted isomerization can occur, generating product **4** via the enolization of oxonium **IC** and **ID**.

3. CONCLUSIONS

In conclusion, we have developed a novel method for the synthesis of penta-substituted aromatic compounds starting from diazo-cumulated allenyl ketones based on the domino cyclopropanation/rearrangement/isomerization strategy. Other noteworthy features of the cyclization reactions include excellent chemoselectivity, mild reaction conditions, and simple manipulation.

4. EXPERIMENTAL SECTION

4.1. General. All reactions were carried out under a nitrogen atmosphere. Dry DCM, DCE, and toluene were obtained by treatment with CaH_2 before use; THF and 1,4-dioxane were treated with sodium and distilled prior to use. For chromatographic purification, 200−300 mesh silica gel (Qingdao, China) was employed. ^{1}H NMR spectra were recorded on a Bruker ARX 500 MHz spectrometer in chloroform-*d*, and all signals are reported in ppm with the internal chloroform signal at 7.26 ppm. The data are being reported as (s = singlet, d = doublet, t = triplet, q = quadruplet, m = multiplet or unresolved, br = broad signal). ^{13}C NMR spectra were recorded with ^{1}H-decoupling on a Bruker ARX 500 MHz spectrometer in chloroform-*d*, and all signals are reported in ppm with the internal chloroform signal at 77.0 ppm. HRMS were

https://dx.doi.org/10.1021/acs.joc.9b03093
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performed on a Thermo Scientific LTQ Qrbitrap XL mass spectrometer (ESI). Infrared spectra (IR) were recorded on a Nicolet Nexus TT-IR spectrophotometer and are reported as wavelength numbers (cm^{-1}). Infrared spectra were recorded by preparing a KBr pellet containing the title compound. Melting points were measured on a YRT-3 melting point apparatus using open glass capillaries, and the data are uncorrected. X-ray data were taken on a Rigaku SuperNova diffractometer equipped with an EOS S2 CCD detector. For thin-layer chromatography (TLC) analysis throughout this work, Merck 25 TLC aluminium sheets (Silica gel 60 GF254, 0.25 mm) were used. Unless otherwise noted, materials obtained from commercial suppliers were used without further purifications.

4.2. General Procedure for the Preparation of Diazo Cumulated Allenyl Ketones 1.[20] $TiCl_4$ (10 mmol, 1.1 mL) and Et_3N (10 mmol, 1 g) were dropwise syringed into a vigorous stirring solution of β-keto α-diazo carbonyl compounds (10 mmol, 1.0 equiv) in anhydrous CH_2Cl_2 (30 mL) at −78 °C under a nitrogen atmosphere. The dark-red mixture was stirred at −78 °C for 1 h. Lewis acid (10 mmol, 1.0 equiv) and α-alkynyl α,β-unsaturated ketones (10 mmol, 1.0 equiv) in anhydrous CH_2Cl_2 (10 mL) were syringed. The reaction mixture was stirred for another 10 h and then quenched with saturated aqueous NH_4Cl (30 mL). The organic layer was separated, washed with saturated aqueous $NaHCO_3$ (30 mL) and NaCl (30 mL) in turn, dried over anhydrous $MgSO_4$, and then concentrated in vacuum. The residue was purified by column chromatography using EA/PE (1:20) as the eluent to afford compounds **1**.

*4.2.1. Ethyl 6-Acetyl-2-diazo-3-oxo-5,8-diphenylocta-6,7-dienoate (**1aa**).* It is obtained as yellow oil (0.78 g, 43%), R_f = 0.30 (ethyl acetate/petroleum ether = 1:10). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.32 (t, J = 8.0 Hz, 4H), 7.27 (d, J = 7.0 Hz, 1H), 7.23 (t, J = 7.5 Hz, 2H), 7.19−7.15 (m, 3H), 6.78 (d, J = 1.0 Hz, 1H), 4.62 (t, J = 7.5 Hz, 1H), 4.23 (q, J = 7.0 Hz, 2H), 3.37 (d, J = 7.5 Hz, 2H), 2.24 (s, 3H), 1.27 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 214.4, 196.4, 190.3, 161.3, 142.5, 131.8, 129.0, 128.4, 128.2, 127.9, 127.2, 126.8, 116.7, 101.4, 61.5, 45.2, 38.3, 27.5, 14.3. IR (KBr): 2926, 2134, 1716, 1682, 1373, 1300, 1028, 746, 695 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + Na]^+$ calcd for $C_{24}H_{22}N_2NaO_4^+$, 425.1472; found, 425.1470.

*4.2.2. Ethyl 6-Acetyl-2-diazo-3-oxo-5-phenyl-8-(p-tolyl)octa-6,7-dienoate (**1ba**).* It is obtained as yellow oil (0.87 g, 50%), R_f = 0.30 (ethyl acetate/petroleum ether = 1:10). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.32 (d, J = 7.5 Hz, 2H), 7.24 (t, J = 7.0 Hz, 2H), 7.17 (t, J = 7.5 Hz, 1H), 7.13 (d, J = 8.0 Hz, 2H), 7.08 (d, J = 8.0 Hz, 2H), 6.75 (s, 1H), 4.61 (t, J = 7.5 Hz, 1H), 4.24 (q, J = 7.0 Hz, 2H), 3.36 (d, J = 7.5 Hz, 2H), 2.35 (s, 3H), 2.23 (s, 3H), 1.27 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 214.3, 196.5, 190.4, 161.3, 142.6, 138.3, 129.7, 128.7, 128.4, 128.0, 127.1, 126.7, 116.7, 116.6, 101.2, 61.5, 45.3, 38.3, 27.5, 21.3, 14.3. IR (KBr): 2922, 2136, 1717, 1682, 1302, 1129, 702 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + Na]^+$ calcd for $C_{25}H_{24}N_2NaO_4^+$, 439.1628; found, 439.1633.

*4.2.3. Ethyl 6-Acetyl-2-diazo-8-(4-methoxyphenyl)-3-oxo-5-phenylocta-6,7-dienoate (**1ca**).* It is obtained as yellow oil (0.90 g, 52%), R_f = 0.30 (ethyl acetate/petroleum ether = 1:10). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.29 (d, J = 7.0 Hz, 2H), 7.22 (s, 1H), 7.19−7.18 (m, 1H), 7.16−7.13 (m, 2H), 6.83 (d, J = 8.5 Hz, 2H), 6.59 (d, J = 8.5 Hz, 2H), 4.67 (t, J = 10.0 Hz, 1H), 4.26 (q, J = 7.0 Hz, 2H), 3.76 (s, 3H), 1.87 (s, 3H), 1.29 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 193.6, 160.9, 159.9, 136.6, 132.9, 131.9, 128.9, 128.2, 128.0, 115.4, 114.1, 98.4, 86.6, 75.6, 61.6, 55.3, 47.9, 29.0, 14.2. IR (KBr): 2136, 1716, 1681, 1512, 1302, 1250, 1030, 847, 703 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + Na]^+$ calcd for $C_{25}H_{24}N_2NaO_5^+$, 455.1577; found, 455.1582.

*4.2.4. Ethyl 6-Acetyl-8-(4-bromophenyl)-2-diazo-3-oxo-5-phenylocta-6,7-dienoate (**1da**).* It is obtained as yellow oil (1.49 g, 50%), R_f = 0.35 (ethyl acetate/petroleum ether = 1:10). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.41 (d, J = 8.5 Hz, 2H), 7.29 (d, J = 7.5 Hz, 2H), 7.23 (t, J = 7.0 Hz, 2H), 7.18−7.15 (m, 1H), 7.00 (d, J = 8.5 Hz, 2H), 6.71 (d, J = 1.0 Hz, 1H), 4.62 (t, J = 7.0 Hz, 1H), 4.24 (q, J = 7.0 Hz, 2H), 3.41 (dd, J = 7.0, 17.0 Hz, 1H), 3.34 (dd, J = 8.5, 17.0 Hz, 1H), 2.22 (s, 3H), 1.28 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 214.5, 196.0, 190.2, 161.3, 142.4, 132.1, 130.8, 128.6, 128.52, 127.8, 126.9, 122.1, 117.1, 100.6, 61.5, 45.0, 38.2, 27.6, 14.3. IR (KBr): 2136, 1715, 1683, 1302, 1011, 846, 702 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + Na]^+$ calcd for $C_{24}H_{21}BrN_2NaO_4^+$, 503.0577; found, 503.0577.

*4.2.5. Ethyl 6-Acetyl-8-(4-chlorophenyl)-2-diazo-3-oxo-5-phenylocta-6,7-dienoate (**1ea**).* It is obtained as yellow oil (1.50 g, 53%), R_f = 0.35 (ethyl acetate/petroleum ether = 1:10). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.23−7.20 (m, 3H), 7.19−7.17 (m, 2H), 7.16−7.14 (m, 1H), 7.12−7.08 (m, 1H), 6.99 (d, J = 8.5 Hz, 2H), 6.65 (d, J = 1.5 Hz, 1H), 4.55 (t, J = 7.5 Hz, 1H), 4.18 (q, J = 7.0 Hz, 2H), 3.34 (dd, J = 7.0, 17.5 Hz, 1H), 3.27 (dd, J = 8.5, 17.0 Hz, 1H), 2.16 (s, 3H), 1.21 (t, J = 7.5 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 214.5, 196.0, 190.2, 161.3, 142.4, 134.0, 130.3, 129.2, 128.5, 128.3, 127.8, 126.8, 117.0, 100.4, 61.5, 45.1, 38.2, 27.5, 14.3. IR (KBr): 2137, 1716, 1684, 1302, 1029, 848, 702 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + Na]^+$ calcd for $C_{24}H_{21}ClN_2NaO_4^+$, 459.1082; found, 459.1086.

*4.2.6. Ethyl 6-Acetyl-2-diazo-8-(4-fluorophenyl)-3-oxo-5-phenylocta-6,7-dienoate (**1fa**).* It is obtained as yellow oil (1.71 g, 58%), R_f = 0.35 (ethyl acetate/petroleum ether = 1:10). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.22 (d, J = 7.5 Hz, 2H), 7.16 (d, J = 7.0 Hz, 2H), 7.09 (t, J = 7.0 Hz, 1H), 7.05−7.02 (m, 2H), 6.92 (t, J = 8.5 Hz, 2H), 6.66 (d, J = 1.5 Hz, 1H), 4.55 (t, J = 7.0 Hz, 1H), 4.18 (q, J = 7.5 Hz, 2H), 3.33 (dd, J = 7.0, 17.0 Hz, 1H), 3.27 (dd, J = 8.5, 17.0 Hz, 1H), 2.16 (s, 3H), 1.21 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 214.1 (d, $^8J_{C(=O)-F}$ = 2.9 Hz), 196.2, 190.2, 162.5 (d, $^1J_{C-F}$ = 248.3 Hz), 161.2, 142.4, 128.7 (d, $^3J_{C-F}$ = 8.0 Hz), 128.4, 127.8, 127.7 (d, $^4J_{C-F}$ = 3.4 Hz), 126.7, 116.9, 116.0 (d, $^2J_{C-F}$ = 21.9 Hz), 100.3, 61.4, 45.0, 38.1, 27.4, 14.2. IR (KBr): 2136, 1715, 1682, 1509, 1226, 1029, 850 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + Na]^+$ calcd for $C_{24}H_{21}FN_2NaO_4^+$, 443.1378; found, 443.1383.

*4.2.7. Ethyl 6-Acetyl-2-diazo-3-oxo-5-phenyl-8-(thiophen-3-yl)-octa-6,7-dienoate (**1ga**).* It is obtained as yellow oil (0.86 g, 45%), R_f = 0.35 (ethyl acetate/petroleum ether = 1:10). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.32 (d, J = 7.5 Hz, 2H), 7.28−7.23 (m, 3H), 7.18 (d, J = 7.0 Hz, 1H), 7.15 (d, J = 2.0 Hz, 1H), 6.86 (d, J = 4.0 Hz, 2H), 4.61 (t, J = 7.5 Hz, 1H), 4.24 (q, J = 7.5 Hz, 2H), 3.37 (d, J = 7.5 Hz, 2H), 2.23 (s, 3H), 1.28 (t, J = 7.5 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 214.7, 196.3, 190.3, 161.3, 142.6, 132.4, 128.5, 127.9, 126.8, 126.7, 126.0, 123.0, 115.9, 95.9, 61.5, 45.2, 38.1, 27.5, 14.3. IR (KBr): 2985, 2136, 1716, 1681, 1302, 1130, 1028, 703 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + Na]^+$ calcd for $C_{22}H_{20}N_2NaO_4S^+$, 431.1036; found, 431.1042.

*4.2.8. Ethyl 6-Acetyl-2-diazo-3-oxo-5-phenyldodeca-6,7-dienoate (**1ha**).* It is obtained as yellow oil (0.64 g, 48%), R_f = 0.30 (ethyl acetate/petroleum ether = 1:10). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.27−7.26 (m, 2H), 7.24−7.22 (m, 2H), 7.17−7.14 (m, 1H), 5.74 (td, J = 7.0 Hz, 1H), 4.55 (td, J = 7.0 Hz, 1H), 4.28 (q, J = 7.0 Hz, 2H), 3.30 (s, 1H), 3.29 (d, J = 1.5 Hz, 1H), 2.21 (s, 3H), 2.17−2.12 (m,2H), 1.39−1.33 (m, 4H), 1.30 (t, J = 7.0 Hz, 3H), 0.90 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 211.7, 197.4, 190.5, 161.3, 142.8, 128.2, 127.8, 126.5, 113.2, 98.3, 61.4, 45.2, 37.5, 31.2, 28.1, 27.2, 22.3, 14.3, 13.8. IR (KBr): 2135, 1716, 1678, 1302, 1128, 1049, 701 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + Na]^+$ calcd for $C_{22}H_{26}N_2NaO_4^+$, 405.1785; found, 405.1796.

*4.2.9. Ethyl 6-Acetyl-2-diazo-9,9-dimethyl-3-oxo-5-phenyldeca-6,7-dienoate (**1ia**).* It is obtained as yellow oil (0.66 g, 43%), R_f = 0.33 (ethyl acetate/petroleum ether = 1:10). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.25−7.20 (m, 4H), 7.14−7.11 (m, 1H), 5.64 (d, J = 2.0 Hz, 1H), 4.52 (dd, J = 2.0, 7.5 Hz, 1H), 4.25 (q, J = 7.5 Hz, 2H), 3.33 (dd, J = 7.5, 17.0 Hz, 1H), 3.26 (dd, J = 8.0, 17.5 Hz, 1H), 2.19 (s, 3H), 1.28 (t, J = 7.0 Hz, 3H), 0.97 (s, 9H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 208.4, 196.1, 189.1, 160.0, 141.3, 126.9, 126.7, 125.1, 113.6, 108.1, 60.0, 43.5, 36.4, 32.0, 28.5, 25.9, 13.0. IR (KBr): 2962, 2135, 1717, 1678, 1302, 1129, 703 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + Na]^+$ calcd for $C_{22}H_{26}N_2NaO_4^+$, 405.1785; found, 405.1795.

*4.2.10. Ethyl 6-Acetyl-2-diazo-3-oxo-5-phenylocta-6,7-dienoate (**1ja**).* It is obtained as yellow oil (0.63 g, 45%), R_f = 0.35 (ethyl acetate/petroleum ether = 1:10). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.29−7.27 (m, 2H), 7.24 (t, J = 7.5 Hz, 2H), 7.15 (t, J = 7.0 Hz, 1H), 5.40−5.33 (m, 2H), 4.51−4.48 (m, 1H), 4.27 (q, J = 7.0 Hz, 2H), 3.37−3.28 (m, 2H), 2.23 (s, 3H), 1.30 (t, J = 7.5 Hz, 3H). ^{13}C {1H} NMR (125.8

https://dx.doi.org/10.1021/acs.joc.9b03093
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MHz, $CDCl_3$, ppm): δ 216.1, 196.7, 190.2, 161.3, 142.6, 128.3, 127.8, 126.6, 112.4, 82.2, 61.4, 45.2, 36.9, 27.3, 14.3. IR (KBr): 2985, 2137, 1933, 1717, 1679, 1303, 1029, 704 cm^{-1}. HRMS (ESI-TOF) *m/z*: $[M + Na]^+$ calcd for $C_{18}H_{18}N_2NaO_4^+$, 349.1159; found, 349.1169.

*4.2.11. 2-Diazo-1,5-diphenyl-6-(2-phenylvinylidene)octane-1,3,7-trione (**1ab**).* It is obtained as yellow oil (0.87 g, 40%), R_f = 0.34 (ethyl acetate/petroleum ether = 1:8). ^1H NMR (500 MHz, $CDCl_3$, ppm): δ 7.56−7.52 (m, 3H), 7.44 (t, *J* = 7.5 Hz, 2H), 7.35−7.31 (m, 3H), 7.30−7.27 (m, 2H), 7.25−7.22 (m, 2H), 7.18−7.15 (m, 3H), 6.79 (d, *J* = 1.5 Hz, 1H), 4.66 (t, *J* = 7.0 Hz, 1H), 3.53 (dd, *J* = 7.5, 17.5 Hz, 1H), 3.41 (dd, *J* = 8.0, 17.0 Hz, 1H), 2.25 (s, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 214.5, 196.4, 191.1, 185.0, 142.4, 137.4, 132.7, 131.7, 129.0, 128.9, 128.5, 128.3, 128.0, 127.3, 127.2, 126.8, 116.6, 101.4, 83.6, 46.3, 38.5, 27.5. IR (KBr): 2139, 1716, 1374, 1296, 1030, 754, 694 cm^{-1}. HRMS (ESI-TOF) *m/z*: $[M + H]^+$ calcd for $C_{28}H_{22}N_2NaO_3^+$, 457.1523; found, 457.1531.

*4.2.12. 2-Diazo-1,5-diphenyl-6-(2-(p-tolyl)vinylidene)octane-1,3,7-trione (**1bb**).* It is obtained as yellow oil (1.88 g, 42%), R_f = 0.30 (ethyl acetate/petroleum ether = 1:8). ^1H NMR (500 MHz, $CDCl_3$, ppm): δ 7.57−7.53 (m, 3H), 7.45 (t, *J* = 7.0 Hz, 2H), 7.35 (d, *J* = 7.5 Hz, 2H), 7.27−7.24 (m, 3H), 7.18 (t, *J* = 7.5 Hz, 1H), 7.13 (d, *J* = 8.0 Hz, 2H), 7.08 (d, *J* = 8.0 Hz, 2H), 6.78 (d, *J* = 1.5 Hz, 1H), 4.67 (t, *J* = 7.5 Hz, 1H), 3.53 (dd, *J* = 7.0, 17.0 Hz, 1H), 3.42 (dd, *J* = 8.0, 17.0 Hz, 1H), 2.35 (s, 3H), 2.25 (s, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 214.4, 196.6, 191.1, 185.0, 142.4, 138.3, 137.4, 132.7, 129.8, 128.9, 128.6, 128.5, 128.0, 127.3, 127.2, 126.8, 116.6, 101.3, 83.6, 46.4, 38.5, 27.5, 21.3. IR (KBr): 2120, 1679, 1646, 1303, 1194, 703 cm^{-1}. HRMS (ESI-TOF) *m/z*: $[M + Na]^+$ calcd for $C_{29}H_{24}N_2NaO_3^+$, 471.1679; found, 471.1697.

*4.2.13. 2-Diazo-6-(2-(4-fluorophenyl)vinylidene)-1,5-diphenyloctane-1,3,7-trione (**1cb**).* It is obtained as yellow oil (0.77 g, 46%), R_f = 0.35 (ethyl acetate/petroleum ether = 1:8). ^1H NMR (500 MHz, $CDCl_3$, ppm): δ 7.58−7.54 (m, 3H), 7.46 (t, *J* = 8.0 Hz, 2H), 7.33 (d, *J* = 7.5 Hz, 2H), 7.24 (t, *J* = 7.5 Hz, 2H), 7.18 (t, *J* = 7.0 Hz, 1H), 7.12−7.09 (m, 2H), 6.99 (t, *J* = 8.5 Hz, 2H), 6.78 (s, 1H), 4.68 (t, *J* = 7.5 Hz, 1H), 3.57 (dd, *J* = 7.5, 17.5 Hz, 1H), 3.43 (dd, *J* = 8.0, 17.0 Hz, 1H), 2.25 (s, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 214.2 (d, $^8J_{c(=o)-F}$ = 2.3 Hz), 196.3, 191.1, 185.2, 162.3 (d, $^1J_{C-F}$ = 248.5 Hz), 142.4, 137.3, 132.8, 129.0, 128.8 (d, $^3J_{C-F}$ = 8.2 Hz), 128.5, 128.0, 127.6 (d, $^4J_{C-F}$ = 2.1 Hz), 127.3, 126.9, 116.9, 116.1 (d, $^2J_{C-F}$ = 21.9 Hz), 100.5, 83.6, 46.2, 38.4, 27.5. IR (KBr): 2120, 1680, 1646, 1508, 1227, 849, 704 cm^{-1}. HRMS (ESI-TOF) *m/z*: $[M + Na]^+$ calcd for $C_{28}H_{21}FN_2NaO_3^+$, 475.1428; found, 475.1444.

*4.2.14. 2-Diazo-1,5-diphenyl-6-(2-(thiophen-3-yl)vinylidene)-octane-1,3,7-trione (**1db**).* It is obtained as yellow oil (2.20 g, 50%), R_f = 0.30 (ethyl acetate/petroleum ether = 1:8). ^1H NMR (500 MHz, $CDCl_3$, ppm): δ 7.56−7.52 (m, 3H), 7.43 (t, *J* = 8.0 Hz, 2H), 7.34 (d, *J* = 7.5 Hz, 2H), 7.26−7.24 (m, 2H), 7.23 (s, 1H), 7.17 (t, *J* = 7.0 Hz, 1H), 7.14 (d, *J* = 2.5 Hz, 1H), 6.88 (s, 1H), 6.84 (d, *J* = 5.0 Hz, 1H), 4.66 (t, *J* = 8.0 Hz, 1H), 3.53 (dd, *J* = 7.0, 17.5 Hz, 1H), 3.41 (dd, *J* = 8.0, 17.0 Hz, 1H), 2.23 (s, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 214.8, 196.4, 191.1, 185.1, 142.5, 137.3, 132.7, 132.3, 129.0, 128.5, 128.0, 127.4, 126.8, 126.8, 126.0, 123.1, 115.9, 96.0, 83.7, 46.3, 38.4, 27.6. IR (KBr): 2121, 1679, 1645, 1304, 1226, 1077, 704 cm^{-1}. HRMS (ESI-TOF) *m/z*: $[M + H]^+$ calcd for $C_{26}H_{20}N_2NaO_3S^+$, 463.1087; found, 463.1100.

*4.2.15. 2-Diazo-6-(hex-1-en-1-ylidene)-1,5-diphenyloctane-1,3,7-trione (**1eb**).* It is obtained as yellow oil (1.95 g, 47%), R_f = 0.30 (ethyl acetate/petroleum ether = 1:8). ^1H NMR (500 MHz, $CDCl_3$, ppm): δ 7.58−7.54 (m, 3H), 7.48−7.46 (m, 2H), 7.30−7.26 (m, 2H), 7.18−7.16 (m, 1H), 5.76 (t, *J* = 7.5 Hz, 1H), 4.56 (t, *J* = 7.5 Hz, 1H), 3.47 (dd, *J* = 7.0, 17.0 Hz, 1H), 3.36 (dd, *J* = 7.5, 17.0 Hz, 1H), 2.22 (s, 3H), 2.15−2.13 (m, 2H), 1.53−1.31 (m, 4H), 0.87 (t, *J* = 7.5 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 211.8, 197.5, 191.2, 185.0, 142.7, 137.4, 132.7, 128.9, 128.3, 127.9, 127.4, 126.6, 113.2, 98.4, 83.4, 46.3, 37.8, 31.2, 28.1, 27.3, 22.4, 13.9. IR (KBr): 2930, 2119, 1646, 1450, 1304, 1193, 889, 704 cm^{-1}. HRMS (ESI-TOF) *m/z*: $[M + Na]^+$ calcd for $C_{26}H_{26}N_2NaO_3^+$, 437.1836; found, 437.1846.

*4.2.16. 2-Diazo-6-(3,3-dimethylbut-1-en-1-ylidene)-1,5-diphenyloctane-1,3,7-trione (**1fb**).* It is obtained as yellow oil (0.46 g, 44%), R_f = 0.30 (ethyl acetate/petroleum ether = 1:8). ^1H NMR (500 MHz, $CDCl_3$, ppm): δ 7.57−7.52 (m, 3H), 7.44 (t, *J* = 7.5 Hz, 2H), 7.28 (d, *J* = 7.5 Hz, 2H), 7.24 (t, *J* = 7.5 Hz, 2H), 7.15 (t, *J* = 7.0 Hz, 1H), 5.68 (d, *J* = 2.0 Hz, 1H), 4.59 (td, *J* = 7.5 Hz, 1H), 3.52 (dd, *J* = 7.5, 17.0 Hz, 1H), 3.35 (dd, *J* = 7.0, 17.0 Hz, 1H), 2.22 (s, 3H), 0.98 (s, 9H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 209.8, 197.6, 191.2, 185.0, 142.5, 137.4, 132.7, 128.9, 128.2, 128.1, 127.3, 126.6, 114.9, 109.4, 83.4, 45.9, 38.0, 33.4, 29.8, 27.3. IR (KBr): 2962, 2117, 1943, 1302, 1030, 893, 704 cm^{-1}. HRMS (ESI-TOF) *m/z*: $[M + Na]^+$ calcd for $C_{26}H_{26}N_2NaO_3^+$, 437.1836; found, 437.1846.

*4.2.17. 2-Diazo-1,5-diphenyl-6-vinylideneoctane-1,3,7-trione (**1gb**).* It is obtained as a yellow solid (0.68 g, 42%), mp 118−120 °C, R_f = 0.30 (ethyl acetate/petroleum ether = 1:8). ^1H NMR (500 MHz, $CDCl_3$, ppm): δ 7.55 (t, *J* = 7.0 Hz, 1H), 7.48−7.42 (m, 4H), 7.28 (t, *J* = 7.5 Hz, 3H), 7.26−7.24 (m, 1H), 7.18 (t, *J* = 7.0 Hz, 1H), 5.62 (s, 1H), 5.56 (s, 1H), 4.06−4.01 (m, 1H), 3.59 (dd, *J* = 10.5, 17.0 Hz, 1H), 3.20 (dd, *J* = 3.0, 17.0 Hz, 1H), 1.99 (s, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 203.3, 191.2, 185.0, 140.9, 137.6, 137.2, 132.7, 129.0, 128.9, 128.6, 128.5, 127.3, 127.0, 118.5, 83.8, 44.3, 40.8, 29.6. IR (KBr): 2121, 1723, 1645, 1302, 1192, 894, 704 cm^{-1}. HRMS (ESI-TOF) *m/z*: $[M + Na]^+$ calcd for $C_{22}H_{18}N_2NaO_3^+$, 381.1210; found, 381.1216.

4.3. General Procedure for the Synthesis of Diazo Trisubstituted Furans 2. Ethyl 6-acetyl-2-diazo-3-oxo-5,8-diphenylocta-6,7-dienoate **1aa** (0.40 g, 0.1 mmol) in DCE (3 mL) was syringed dropwise into a vigorous stirring solution of CF_3CO_2Ag (6.9 mg, 0.02 mmol) in DCE (5 mL) in an oil bath at 60 °C for 30 min. After the completion of the reaction (monitored by TLC), the mixture was concentrated under reduced pressure. The residue was purified by column chromatography on silica gel with a suitable mixture of PE and EA to provide the corresponding product **2aa** (86.7 mg) in 82% yield.

*4.3.1. Ethyl 2-Diazo-5-(2-methyl-5-phenylfuran-3-yl)-3-oxo-5-phenylpentanoate (**2aa**).* It is obtained as a yellow solid (66.0 mg, 82%), mp 125−127 °C. R_f = 0.60 (ethyl acetate/petroleum ether = 1:6). ^1H NMR (500 MHz, $CDCl_3$, ppm): δ 7.59 (d, *J* = 7.5 Hz, 2H), 7.34−7.27 (m, 6H), 7.19 (t, *J* = 7.0 Hz, 2H), 6.60 (s, 1H), 4.56 (t, *J* = 7.5 Hz, 1H), 4.30 (q, *J* = 7.0 Hz, 2H), 3.67 (dd, *J* = 8.5, 16.5 Hz, 1H), 3.50 (dd, *J* = 7.0, 17.0 Hz, 1H), 2.32 (s, 3H), 1.32 (t, *J* = 7.2 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 190.8, 161.4, 151.4, 147.6, 143.7, 131.0, 128.6, 128.5, 127.5, 126.7, 126.4, 123.4, 123.2, 105.6, 61.4, 45.5, 37.0, 14.3, 11.9. IR (KBr): 2925, 1714, 1651, 1303, 1049, 757, 697 cm^{-1}. HRMS (ESI-TOF) *m/z*: $[M + H]^+$ calcd for $C_{24}H_{23}N_2O_4^+$, 403.1652; found, 403.1664.

*4.3.2. Ethyl 2-Diazo-5-(2-methyl-5-(p-tolyl)furan-3-yl)-3-oxo-5-phenylpentanoate (**2ba**).* It is obtained as yellow oil (65.0 mg, 78%). R_f = 0.65 (ethyl acetate/petroleum ether = 1:6). ^1H NMR (500 MHz, $CDCl_3$, ppm): δ 7.50 (d, *J* = 8.0 Hz, 2H), 7.35−7.34 (m, 2H), 7.30 (t, *J* = 7.5 Hz, 2H), 7.20 (t, *J* = 7.0 Hz, 1H), 7.15 (d, *J* = 8.0 Hz, 2H), 6.56 (s, 1H), 4.57 (t, *J* = 7.0 Hz, 1H), 4.30 (q, *J* = 7.0 Hz, 2H), 3.68 (dd, *J* = 8.5, 17.0 Hz, 1H), 3.52 (dd, *J* = 6.5, 17.0 Hz, 1H), 2.35 (s, 3H), 2.34 (s, 3H), 1.32 (t, *J* = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 190.8, 161.4, 151.6, 147.1, 143.8, 136.5, 129.2, 128.6, 128.4, 127.5, 126.4, 123.3, 123.2, 104.9, 76.4, 61.5, 45.6, 37.0, 21.25, 14.4, 11.9. IR (KBr): 2918, 2118, 1644, 1302, 1191, 780, 704 cm^{-1}. HRMS (ESI-TOF) *m/z*: $[M + H]^+$ calcd for $C_{25}H_{25}N_2O_4^+$, 417.1809; found, 417.1811.

*4.3.3. Ethyl-2-diazo-5-(5-(4-methoxyphenyl)-2-methylfuran-3-yl)-3-oxo-5-phenylpentanoate (**2ca**).* It is obtained as yellow oil (69.2 mg, 80%). R_f = 0.65 (ethyl acetate/petroleum ether = 1:6). ^1H NMR (500 MHz, $CDCl_3$, ppm): δ 7.53 (d, *J* = 8.5 Hz, 2H), 7.34−7.33 (m, 2H), 7.30 (t, *J* = 7.7 Hz, 2H), 7.19 (t, *J* = 7.2 Hz, 1H), 6.88 (d, *J* = 9.0 Hz, 2H), 6.48 (s, 1H), 4.56 (t, *J* = 7.5 Hz, 1H), 4.30 (q, *J* = 7.0 Hz, 2H), 3.81 (s, 3H), 3.67 (dd, *J* = 8.5, 17.0 Hz, 1H), 3.51 (dd, *J* = 6.5, 16.5 Hz, 1H), 2.32 (s, 3H), 1.32 (t, *J* = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 190.8, 161.4, 158.6, 151.4, 146.7, 143.8, 128.5, 127.5, 126.4, 124.7, 124.3, 123.2, 114.0, 104.0, 76.4, 61.4, 55.3, 45.5, 37.0, 14.4, 11.9. IR (KBr): 2137, 1713, 1252, 1029, 749, 700 cm^{-1}. HRMS (ESI-TOF) *m/z*: $[M + Na]^+$ calcd for $C_{25}H_{24}N_2NaO_5^+$, 455.1577; found, 455.1576.

https://dx.doi.org/10.1021/acs.joc.9b03093
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*4.3.4. Ethyl-5-(5-(4-bromophenyl)-2-methylfuran-3-yl)-2-diazo-3-oxo-5-phenylpentanoate (**2da**).* It is obtained as a yellow solid (78.0 mg, 81%), mp 131−133 °C. R_f = 0.70 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.45 (s, 4H), 7.34−7.29 (m, 4H), 7.20 (t, *J* = 7.7 Hz, 1H), 6.62 (s, 1H), 4.56 (dd, *J* = 6.5, 8.5 Hz, 1H), 4.30 (q, *J* = 7.5 Hz, 2H), 3.66 (dd, *J* = 8.5, 17.0 Hz, 1H), 3.52 (dd, *J* = 6.5, 16.5 Hz, 1H), 2.33 (s, 3H), 1.32 (t, *J* = 7.3 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 190.6, 161.2, 150.3, 147.9, 143.5, 131.5, 129.9, 128.5, 127.3, 126.4, 124.6, 123.6, 120.2, 106.1, 76.3, 61.4, 45.4, 36.8, 14.3, 11.9. IR (KBr): 2982, 2136, 1714, 1655, 1305, 1050, 810, 703 cm^{-1}. HRMS (ESI-TOF) *m*/*z*: [M + H]$^+$ calcd for $C_{24}H_{22}BrN_2O_4^+$, 481.0757; found, 481.0757.

*4.3.5. Ethyl-5-(5-(4-chlorophenyl)-2-methylfuran-3-yl)-2-diazo-3-oxo-5-phenylpentanoate (**2ea**).* It is obtained as a yellow solid (69.9 mg, 80%), mp 126−128 °C. R_f = 0.65 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.52 (d, *J* = 8.5 Hz, 2H), 7.35 (t, *J* = 7.5 Hz, 2H), 7.32−7.29 (m, 3H), 7.21 (t, *J* = 7.2 Hz, 1H), 6.63 (s, 1H), 4.59 (t, *J* = 7.5 Hz, 1H), 4.30 (q, *J* = 7.0 Hz, 2H), 3.69 (dd, *J* = 8.5, 16.5 Hz, 1H), 3.55 (dd, *J* = 6.5, 16.5 Hz, 1H), 2.35 (s, 3H), 1.32 (t, *J* = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 190.7, 161.4, 150.4, 148.0, 143.6, 132.2, 129.6, 128.7, 128.6, 127.5, 126.5, 124.5, 123.8, 106.1, 76.4, 61.5, 45.5, 37.0, 14.4, 12.0. IR (KBr): 2925, 2137, 1715, 1485, 1305, 1093, 704 cm^{-1}. HRMS (ESI-TOF) *m*/*z*: [M + H]$^+$ calcd for $C_{24}H_{22}ClN_2O_4^+$, 437.1263; found, 437.1260.

*4.3.6. Ethyl-2-diazo-5-(5-(4-fluorophenyl)-2-methylfuran-3-yl)-3-oxo-5-phenylpentanoate (**2fa**).* It is obtained as a yellow solid (65.6 mg, 78%), mp 130−132 °C. R_f = 0.65 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.51−7.48 (m, 2H), 7.32−7.31 (m, 2H), 7.26 (t, *J* = 7.5 Hz, 2H), 7.14 (t, *J* = 7.0 Hz, 1H), 6.96 (t, *J* = 8.5 Hz, 2H), 6.53 (s, 1H), 4.56 (t, *J* = 7.5 Hz, 1H), 4.22 (q, *J* = 7.0 Hz, 2H), 3.65 (dd, *J* = 8.5, 17.0 Hz, 1H), 3.51 (dd, *J* = 6.5, 16.5 Hz, 1H), 2.29 (s, 3H), 1.24 (t, *J* = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 190.6, 161.9 (d, $^1J_{C-F}$ = 246.2 Hz), 161.4, 150.6, 147.6, 143.8, 128.6, 127.5, 126.5, 125.0, 125.0, 123.7, 115.5 (d, $^2J_{C-F}$ = 21.8 Hz), 105.4, 76.3, 61.5, 45.5, 37.0, 14.3, 11.9. IR (KBr): 2983, 2137, 1716, 1655, 1498, 1373, 1305, 702 cm^{-1}. HRMS (ESI-TOF) *m*/*z*: [M + H]$^+$ calcd for $C_{24}H_{22}FN_2O_4^+$, 421.1558; found, 421.1569.

*4.3.7. Ethyl 2-diazo-5-(2-methyl-5-(thiophen-3-yl)furan-3-yl)-3-oxo-5-phenylpentanoate (**2ga**).* It is obtained as a yellow solid (65.4 mg, 80%), mp 122−124 °C. R_f = 0.65 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.26−7.25 (m, 1H), 7.23−7.21 (m, 3H), 7.21−7.20 (m, 1H), 7.18−7.17 (m, 1H), 7.16−7.14 (m, 1H), 7.09 (t, *J* = 7.0 Hz, 1H), 6.32 (s, 1H), 4.46 (t, *J* = 7.5 Hz, 1H), 4.20 (q, *J* = 7.0 Hz, 2H), 3.56 (dd, *J* = 8.5, 16.5 Hz, 1H), 3.40 (dd, *J* = 6.5, 16.5 Hz, 1H), 2.21 (s, 3H), 1.22 (t, *J* = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 190.6, 161.3, 148.5, 146.7, 143.6, 132.8, 128.5, 127.4, 126.3, 125.9, 124.4, 122.9, 117.6, 105.2, 76.3, 61.4, 45.4, 36.8, 14.3, 11.8. IR (KBr): 2136, 1713, 1654, 1373, 1303, 1048, 854, 777 cm^{-1}. HRMS (ESI-TOF) *m*/*z*: [M + H]$^+$ calcd for $C_{22}H_{21}N_2O_4S^+$, 409.1217; found, 409.1219.

*4.3.8. Ethyl 5-(5-Butyl-2-methylfuran-3-yl)-2-diazo-3-oxo-5-phenylpentanoate (**2ha**).* It is obtained as yellow oil (72.7 mg, 95%). R_f = 0.65 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.28−7.26 (m, 3H), 7.25−7.23 (m, 1H), 7.16−7.14 (m, 1H), 5.90 (s, 1H), 4.46 (t, *J* = 7.5 Hz, 1H), 4.27 (q, *J* = 7.5 Hz, 2H), 3.54 (dd, *J* = 8.0, 16.5 Hz, 1H), 3.46 (dd, *J* = 7.0, 16.5 Hz, 1H), 2.49 (t, *J* = 7.5 Hz, 2H), 2.18 (s, 3H), 1.58−1.52 (m, 2H), 1.35−1.32 (m, 2H), 1.30 (t, *J* = 7.0 Hz, 3H), 0.90 (t, *J* = 7.5 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 190.9, 161.4, 153.9, 145.2, 144.2, 128.5, 127.5, 126.2, 121.4, 105.0, 76.3, 61.4, 45.6, 37.2, 30.2, 27.9, 22.4, 14.4, 13.9, 11.7. IR (KBr): 2959, 2136, 1716, 1654, 1303, 1049, 702 cm^{-1}. HRMS (ESI-TOF) *m*/*z*: [M + H]$^+$ calcd for $C_{22}H_{27}N_2O_4^+$, 383.1965; found, 383.1965.

*4.3.9. Ethyl 5-(5-(tert-Butyl)-2-methylfuran-3-yl)-2-diazo-3-oxo-5-phenylpentanoate (**2ia**).* It is obtained as yellow oil (61.2 mg, 80%). R_f = 0.65 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.27−7.24 (m, 4H), 7.14 (s, 1H), 5.85 (s, 1H), 4.45 (t, *J* = 7.5 Hz, 1H), 4.27 (q, *J* = 7.0 Hz, 2H), 3.56−3.46 (m, 2H), 2.18 (s, 3H), 1.30 (t, *J* = 7.0 Hz, 3H), 1.20 (s, 9H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 191.0, 161.6, 161.4, 145.0, 144.2, 128.4, 127.6, 126.2, 121.1, 102.1, 76.3, 61.4, 45.7, 37.3, 32.4, 29.2, 14.4, 11.8. IR (KBr): 2968, 2135, 1717, 1657, 1372, 1303, 1048, 701 cm^{-1}. HRMS (ESI-TOF) *m*/*z*: [M + H]$^+$ calcd for $C_{22}H_{27}N_2O_4^+$, 383.1965; found, 383.1966.

*4.3.10. Ethyl 2-Diazo-5-(2-methylfuran-3-yl)-3-oxo-5-phenylpentanoate (**2ja**).* It is obtained as yellow oil (42.4 mg, 65%). R_f = 0.75 (ethyl acetate/petroleum ether = 1 6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.28−7.23 (m, 4H), 7.18 (d, *J* = 1.5 Hz, 1H), 7.17−7.14 (m, 1H), 6.31 (d, *J* = 2.0 Hz, 1H), 4.52 (t, *J* = 7.5 Hz, 1H), 4.27 (q, *J* = 7.0 Hz, 2H), 3.59 (dd, *J* = 8.5, 16.5 Hz, 1H), 3.47 (dd, *J* = 6.5, 16.5 Hz, 1H), 2.22 (s, 3H), 1.30 (t, *J* = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 190.8, 161.3, 147.6, 143.9, 140.1, 128.5, 127.5, 126.4, 121.1, 110.0, 76.4, 61.5, 45.6, 36.9, 14.4, 11.7. IR (KBr): 2983, 2136, 1716, 1656, 1304, 1132, 1053, 705 cm^{-1}. HRMS (ESI-TOF) *m*/*z*: [M + H]$^+$ calcd for $C_{18}H_{19}N_2O_4^+$, 327.1339; found, 327.1342.

*4.3.11. 2-Diazo-5-(2-methyl-5-phenylfuran-3-yl)-1,5-diphenylpentane-1,3-dione (**2ab**).* It is obtained as a yellow solid (69.5 mg, 80%), mp 136−138 °C. R_f = 0.65 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.61−7.54 (m, 5H), 7.44 (t, *J* = 7.5 Hz, 2H), 7.38 (d, *J* = 7.5 Hz, 2H), 7.33 (t, *J* = 7.5 Hz, 4H), 7.24−7.18 (m, 2H), 6.65 (s, 1H), 4.63 (t, *J* = 7.5 Hz, 1H), 3.64 (dd, *J* = 9.0, 16.5 Hz, 1H), 3.47 (dd, *J* = 6.5, 16.5 Hz, 1H), 2.37 (s, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 191.7, 185.2, 151.5, 147.7, 143.6, 137.3, 132.8, 131.0, 129.0, 128.7, 128.6, 127.6, 127.4, 126.8, 126.6, 123.4, 123.3, 105.7, 83.7, 46.6, 37.4, 12.0. IR (KBr): 2925, 2121, 1643, 1305, 1076, 698 cm^{-1}. HRMS (ESI-TOF) *m*/*z*: [M + Na]$^+$ calcd for $C_{28}H_{22}N_2NaO_3^+$, 457.1523; found, 457.1530.

*4.3.12. 2-Diazo-5-(2-methyl-5-(p-tolyl)furan-3-yl)-1,5-diphenylpentane-1,3-dione (**2bb**).* It is obtained as a yellow solid (71.8 mg, 80%), mp 138−140 °C. R_f = 0.55 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.60−7.57 (m, 2H), 7.56−7.52 (m, 3H), 7.45 (t, *J* = 7.7 Hz, 2H), 7.41 (d, *J* = 7.5 Hz, 2H), 7.35 (t, *J* = 7.5 Hz, 2H), 7.24 (t, *J* = 7.5 Hz, 1H), 7.16 (d, *J* = 8.0 Hz, 2H), 6.63 (s, 1H), 4.65 (t, *J* = 7.0 Hz, 1H), 3.82 (dd, *J* = 8.5, 16.5 Hz, 1H), 3.68 (dd, *J* = 6.5, 16.5 Hz, 1H), 2.38 (s, 3H), 2.36 (s, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 191.7, 185.2, 151.8, 147.2, 143.7, 137.4, 136.6, 132.8, 129.3, 129.0, 128.7, 128.4, 127.6, 127.4, 126.6, 123.3, 123.3, 105.0, 83.7, 46.6, 37.5, 21.31, 12.0. IR (KBr): 3027, 2919, 2118, 1646, 1450, 1305, 1192, 701 cm^{-1}. HRMS (ESI-TOF) *m*/*z*: [M + H]$^+$ calcd for $C_{29}H_{25}N_2O_3^+$, 449.1860; found, 449.1866.

*4.3.13. 2-Diazo-5-(5-(4-fluorophenyl)-2-methylfuran-3-yl)-1,5-diphenylpentane-1,3-dione (**2cb**).* It is obtained as a yellow solid (67.9 mg, 75%), mp 138−140 °C. R_f = 0.55 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.57−7.52 (m, 5H), 7.44 (t, *J* = 7.5 Hz, 2H), 7.36 (d, *J* = 8.0 Hz, 2H), 7.32 (t, *J* = 7.0 Hz, 2H), 7.21 (t, *J* = 7.5 Hz, 1H), 7.01 (t, *J* = 8.5 Hz, 2H), 6.56 (s, 1H), 4.59 (t, *J* = 8.5 Hz, 1H), 3.79 (dd, *J* = 8.5, 16.5 Hz, 1H), 3.63 (dd, *J* = 6.5, 16.5 Hz, 1H), 2.35 (s, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 191.7, 185.1, 162.8, 160.8, 150.7, 147.6, 143.5, 137.3, 132.8, 129.0, 128.7, 127.5, 127.4, 126.6, 125.0 (d, J_{C-F} = 8.0 Hz), 123.4, 115.5 (d, J_{C-F} = 21.9 Hz), 105.3, 83.6, 46.5, 37.4, 11.9. IR (KBr): 2120, 1645, 1497, 1304, 1228, 704 cm^{-1}. HRMS (ESI-TOF) *m*/*z*: [M + Na]$^+$ calcd for $C_{28}H_{21}FN_2NaO_3^+$, 475.1428; found, 475.1439.

*4.3.14. 2-Diazo-5-(2-methyl-5-(thiophen-3-yl)furan-3-yl)-1,5-diphenylpentane-1,3-dione (**2db**).* It is obtained as a yellow solid (69.6 mg, 79%), mp 135−137 °C. R_f = 0.55 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.54 (t, *J* = 8.0 Hz, 3H), 7.44 (t, *J* = 7.5 Hz, 2H), 7.34 (d, *J* = 7.5 Hz, 3H), 7.30 (t, *J* = 7.5 Hz, 2H), 7.25 (t, *J* = 2.0 Hz, 1H), 7.21 (d, *J* = 6.0 Hz, 2H), 5.42−5.33 (m, 2H), 4.58 (t, *J* = 8.0 Hz, 1H), 3.76 (dd, *J* = 8.5, 16.5 Hz, 1H), 3.61 (dd, *J* = 7.0, 16.5 Hz, 1H), 2.32 (s, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 216.2, 196.8, 191.0, 185.0, 142.5, 137.3, 132.8, 129.0, 128.9, 128.6, 128.4, 128.3, 127.9, 127.4, 127.3, 126.7, 112.4, 83.4, 82.3, 46.3, 37.2, 27.3. IR (KBr): 2918, 2118, 1302, 1191, 854, 780, 704 cm^{-1}. HRMS (ESI-TOF) *m*/*z*: [M + H]$^+$ calcd for $C_{26}H_{21}N_2O_3S^+$, 441.1267; found, 441.1263.

*4.3.15. 5-(5-Butyl-2-methylfuran-3-yl)-2-diazo-1,5-diphenylpentane-1,3-dione (**2eb**).* It is obtained as yellow oil (68.0 mg, 82%). R_f = 0.65 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.56 (d, *J* = 7.5 Hz, 3H), 7.48 (t, *J* = 7.5 Hz, 2H), 7.32−

https://dx.doi.org/10.1021/acs.joc.9b03093
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7.28 (m, 4H), 7.18 (t, J = 7.0 Hz, 1H), 5.92 (s, 1H), 4.50 (t, J = 7.5 Hz, 1H), 3.65 (dd, J = 8.5, 16.5 Hz, 1H), 3.58 (dd, J = 7.0, 16.0 Hz, 1H), 2.49 (t, J = 7.5 Hz, 2H), 2.22 (s, 3H), 1.57−1.51 (m, 2H), 1.37−1.30 (m, 2H), 0.90 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 191.8, 185.1, 154.1, 145.3, 144.0, 137.4, 132.8, 128.9, 128.5, 127.5, 127.4, 126.3, 121.3, 105.0, 83.6, 46.6, 37.6, 30.1, 27.9, 22.4, 13.9, 11.7. IR (KBr): 2930, 2117, 1647, 1305, 1192, 702 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{26}H_{27}N_2O_3^+$, 415.2016; found, 415.2016.

*4.3.16. 5-(5-(tert-Butyl)-2-methylfuran-3-yl)-2-diazo-1,5-diphenylpentane-1,3-dione (**2fb**).* It is obtained as a yellow solid (70.5 mg, 85%), mp 118−120 °C. R_f = 0.65 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.56 (d, J = 7.5 Hz, 3H), 7.47 (t, J = 7.0 Hz, 2H), 7.32−7.27 (m, 4H), 7.20−7.17 (m, 1H), 5.88 (s, 1H), 4.49 (t, J = 7.5 Hz, 1H), 3.66 (dd, J = 8.0, 16.5 Hz, 1H), 3.57 (dd, J = 7.5, 16.5 Hz, 1H), 2.21 (s, 3H), 1.21 (s, 9H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 191.8, 185.1, 161.7, 145.0, 144.1, 137.4, 132.7, 128.9, 128.5, 127.6, 127.4, 126.3, 121.0, 102.1, 83.7, 46.8, 37.7, 32.4, 29.2, 11.8. IR (KBr): 2967, 2118, 1647, 1305, 1193, 702 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + Na]^+$ calcd for chemical formula: $C_{26}H_{26}N_2NaO_3^+$, 437.1836; found, 437.1847.

*4.3.17. 2-Diazo-5-(2-methylfuran-3-yl)-1,5-diphenylpentane-1,3-dione (**2gb**).* It is obtained as a yellow solid (56.6 mg, 79%), mp 122−124 °C. R_f = 0.65 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.60 (d, J = 7.0 Hz, 2H), 7.55 (t, J = 7.5 Hz, 1H), 7.46 (t, J = 7.5 Hz, 2H), 7.37 (d, J = 7.5 Hz, 2H), 7.32 (t, J = 7.5 Hz, 2H), 7.24−7.20 (m, 2H), 6.41 (d, J = 1.5 Hz, 1H), 4.64 (t, J = 7.0 Hz, 1H), 3.75 (dd, J = 8.5, 16.5 Hz, 1H), 3.67 (dd, J = 7.0, 16.5 Hz, 1H), 2.30 (s, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 191.6, 185.0, 147.7, 143.9, 140.3, 137.4, 132.8, 129.0, 128.6, 127.6, 127.4, 126.5, 121.2, 110.2, 83.7, 46.8, 37.4, 11.8. IR (KBr): 3028, 2119, 1647, 1447, 1306, 1192, 701 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{22}H_{18}N_2NaO_3^+$, 381.1210; found, 381.1227.

4.4. General Procedure for the Synthesis of Compound 3 and 4. Diazo-trisubstituted furans **2** (0.2 mmol) in DCM (1 mL) were syringed dropwise into a vigorous stirring solution of $Rh_2(OAc)_4$ (8.8 mg, 0.02 mmol) in anhydrous DCM (2 mL) in a water bath at 25 °C for 30 min. After the completion of the reaction (monitored by TLC), the mixture was concentrated under reduced pressure. The crude product was purified by column chromatography on silica gel with a suitable mixture of PE and EA to afford the corresponding product **3** or **4**.

*4.4.1. (Z)-Ethyl-5-methyl-3-oxo-6-(2-oxo-2-phenylethylidene)-1,2,3,6-tetrahydro-[1,1′-biphenyl]-4-carboxylate (**3aa**).* It is obtained as a yellow solid (57.7 mg, 77%), mp 126−128 °C. R_f = 0.30 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.82 (d, J = 7.5 Hz, 2H), 7.57 (t, J = 7.0 Hz, 1H), 7.45 (t, J = 7.5 Hz, 2H), 7.41 (t, J = 7.5 Hz, 2H), 7.32 (t, J = 7.5 Hz, 1H), 7.27 (d, J = 7.0 Hz, 2H), 6.36 (d, J = 1.0 Hz, 1H), 4.30 (q, J = 7.5 Hz, 2H), 4.23 (dd, J = 4.5, 10.5 Hz, 1H), 3.10 (dd, J = 10.5, 17.0 Hz, 1H), 2.96 (dd, J = 5.0, 16.5 Hz, 1H), 1.94 (s, 3H), 1.31 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 193.6, 193.2, 165.7, 152.0, 149.1, 138.7, 136.5, 135.5, 133.8, 130.7, 129.0, 128.8, 128.7, 128.0, 127.8, 61.4, 49.3, 43.1, 20.3, 14.0. IR (KBr): 2983, 1660, 1312, 1249, 1210, 995, 757, 670 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{24}H_{23}O_4^+$, 375.1591; found, 375.1594.

*4.4.2. (Z)-Ethyl-6-(2-(4-fluorophenyl)-2-oxoethylidene)-5-methyl-3-oxo-1,2,3,6-tetrahydro-[1,1′-biphenyl]-4-carboxylate (**3fa**).* It is obtained as yellow oil (64.4 mg, 82%). R_f = 0.30 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.87−7.84 (m, 2H), 7.42 (t, J = 7.5 Hz, 2H), 7.34 (t, J = 7.5 Hz, 1H), 7.28 (d, J = 2.5 Hz, 2H), 7.12 (t, J = 8.5 Hz, 2H), 6.34 (d, J = 1.0 Hz, 1H), 4.31 (q, J = 7.5 Hz, 2H), 4.24 (dd, J = 4.5, 10.0 Hz, 1H), 3.10 (dd, J = 10.5, 17.0 Hz, 1H), 2.97 (dd, J = 4.5, 16.5 Hz, 1H), 1.94 (s, 3H), 1.31 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 193.3, 192.0, 166.1 (d, $^1J_{C-F}$ = 256.6 Hz), 165.8, 151.9, 149.6, 138.8, 135.8, 133.1 (d, $^4J_{C-F}$ = 2.9 Hz), 131.5 (d, $^3J_{C-F}$ = 9.6 Hz), 130.2, 129.2, 128.1, 128.0, 116.1 (d, $^2J_{C-F}$ = 22.0 Hz), 61.6, 49.4, 43.1, 20.4, 14.2. IR (KBr): 2983, 1663, 1599, 1240, 704 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{24}H_{22}FO_4^+$, 393.1497; found, 393.1502.

*4.4.3. (Z)-Ethyl-5-methyl-3-oxo-6-(2-oxohexylidene)-1,2,3,6-tetrahydro-[1,1′-biphenyl]-4-carboxylate (**3ha**).* It is obtained as yellow oil (59.5 mg, 84%). R_f = 0.30 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.37 (t, J = 7.5 Hz, 2H), 7.32−7.29 (m, 1H), 7.18 (d, J = 7.0 Hz, 2H), 5.89 (d, J = 1.5 Hz, 1H), 4.31 (q, J = 7.0 Hz, 2H), 4.10 (dd, J = 4.5, 9.5 Hz, 1H), 2.99 (dd, J = 10.0, 17.0 Hz, 1H), 2.88 (dd, J = 5.0, 17.0 Hz, 1H), 2.45 (t, J = 7.0 Hz, 2H), 1.99 (s, 3H), 1.54−1.49 (m, 2H), 1.32 (t, J = 7.0 Hz, 3H), 1.29−1.26 (m, 2H), 0.87 (t, J = 7.5 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 202.3, 193.2, 165.7, 153.1, 149.6, 138.4, 135.4, 130.7, 129.0, 128.0, 127.9, 61.5, 49.8, 43.4, 43.2, 25.9, 22.2, 20.4, 14.2, 13.8. IR (KBr): 2960, 1717, 1461, 1247, 1123, 703 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + Na]^+$ calcd for $C_{22}H_{26}NaO_4^+$, 377.1723; found, 377.1733.

*4.4.4. (Z)-Ethyl-6-(3,3-dimethyl-2-oxobutylidene)-5-methyl-3-oxo-1,2,3,6-tetrahydro-[1,1′-biphenyl]-4-carboxylate (**3ia**).* It is obtained as yellow oil (42.5 mg, 60%). R_f = 0.30 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.38 (t, J = 7.5 Hz, 2H), 7.31 (t, J = 7.5 Hz, 1H), 7.18 (d, J = 8.0 Hz, 2H), 6.16 (s, 1H), 4.31 (q, J = 7.0 Hz, 2H), 4.14 (dd, J = 5.0, 10.0 Hz, 1H), 3.02 (dd, J = 10.0, 17.0 Hz, 1H), 2.91 (dd, J = 5.0, 17.0 Hz, 1H), 1.95 (s, 3H), 1.32 (t, J = 7.0 Hz, 3H), 1.08 (s, 9H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 207.6, 193.4, 165.8, 153.2, 150.5, 138.6, 135.4, 129.0, 128.1, 127.9, 127.8, 61.5, 50.0, 44.0, 43.0, 26.5, 20.2, 14.2. IR (KBr): 2972, 1662, 1372, 1245, 1037, 701 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + Na]^+$ calcd for $C_{22}H_{26}NaO_4^+$, 377.1723; found, 377.1731.

*4.4.5. (Z)-Ethyl-5-methyl-3-oxo-6-(2-oxoethylidene)-1,2,3,6-tetra-hydro-[1,1′-biphenyl]-4-carboxylate (**3ja**).* It is obtained as a yellow solid (48.9 mg, 82%), mp 98−100 °C. R_f = 0.30 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 10.10 (d, J = 7.5 Hz, 1H), 7.38 (t, J = 7.0 Hz, 2H), 7.32 (t, J = 4.0 Hz, 1H), 7.17 (d, J = 7.0 Hz, 2H), 5.82 (d, J = 8.0 Hz, 1H), 4.36 (q, J = 7.0 Hz, 2H), 4.24 (dd, J = 5.5, 9.5 Hz, 1H), 3.06 (dd, J = 9.0, 17.0 Hz, 1H), 2.97 (dd, J = 5.0, 17.0 Hz, 1H), 2.35 (s, 3H), 1.35 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 192.9, 190.7, 165.4, 157.4, 149.9, 137.7, 136.6, 130.9, 129.1, 128.1, 127.9, 61.9, 49.8, 42.6, 23.8, 14.2. IR (KBr): 2983, 1732, 1671, 1234, 1065, 702 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{18}H_{19}O_4^+$, 299.1278; found, 299.1283.

*4.4.6. (Z)-4-Benzoyl-5-methyl-6-(2-oxo-2-phenylethylidene)-1,6-dihydro-[1,1′-biphenyl]-3(2H)-one (**3ab**).* It is obtained as yellow oil (34.1 mg, 42%). R_f = 0.30 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.80 (d, J = 8.0 Hz, 2H), 7.70 (d, J = 7.0 Hz, 2H), 7.60 (t, J = 7.0 Hz, 1H), 7.54 (t, J = 7.5 Hz, 1H), 7.50−7.45 (m, 4H), 7.39 (t, J = 8.5 Hz, 5H), 6.58 (d, J = 1.0 Hz, 1H), 4.38 (dd, J = 4.5, 8.5 Hz, 1H), 3.28 (dd, J = 9.0, 17.0 Hz, 1H), 3.08 (dd, J = 4.5, 17.0 Hz, 1H), 1.80 (s, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 195.7, 195.0, 193.5, 151.8, 150.0, 140.2, 139.0, 136.7, 136.1, 134.0, 133.9, 130.0, 129.2, 129.1, 129.0, 128.9, 128.9, 128.0, 127.9, 49.8, 43.5, 20.4. IR (KBr): 2974, 1679, 1451, 1242, 1365, 1049, 695 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{28}H_{23}O_3^+$, 407.1642; found, 407.1658.

*4.4.7. (Z)-4-Benzoyl-5-methyl-6-(2-oxo-2-(p-tolyl)ethylidene)-1,6-dihydro-[1,1′-biphenyl]-3(2H)-one (**3bb**).* It is obtained as yellow oil (54.7 mg, 65%). R_f = 0.30 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.80 (d, J = 8.0 Hz, 2H), 7.68 (d, J = 7.5 Hz, 2H), 7.53 (t, J = 7.5 Hz, 1H), 7.45 (t, J = 8.0 Hz, 2H), 7.39−7.36 (m, 5H), 7.28 (s, 1H), 7.27 (s, 1H), 6.57 (s, 1H), 4.36 (t, J = 6.5 Hz, 1H), 3.28 (dd, J = 8.5, 16.5 Hz, 1H), 3.07 (dd, J = 4.5, 17.0 Hz, 1H), 2.41 (s, 3H), 1.79 (s, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 195.8, 195.0, 193.3, 151.9, 149.2, 145.1, 140.1, 139.1, 136.1, 134.3, 134.0, 130.4, 129.7, 129.2, 129.1, 129.0, 128.9, 128.0, 127.9, 49.7, 43.5, 21.8, 20.4. IR (KBr): 1680, 1660, 1245, 1178, 1018, 698 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{29}H_{25}O_3^+$, 421.1798; found, 421.1804.

*4.4.8. (Z)-4-Benzoyl-4′-fluoro-5-methyl-6-(2-oxo-2-phenylethylidene)-1,6-dihydro-[1,1′-biphenyl]-3(2H)-one (**3cb**).* It is obtained as yellow oil (67.9 mg, 70%). R_f = 0.30 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.93−7.90 (m, 2H), 7.70 (d, J = 7.0 Hz, 2H), 7.56−7.53 (m, 1H), 7.46 (t, J = 7.5 Hz, 2H), 7.41−7.36 (m, 5H), 7.15 (t, J = 8.5 Hz, 2H), 6.54 (d, J = 1.5 Hz, 1H), 4.37 (dd, J = 5.0, 9.0 Hz, 1H), 3.28 (dd, J = 8.0, 17.0 Hz, 1H), 3.08 (dd, J =

https://dx.doi.org/10.1021/acs.joc.9b03093
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5.0, 17.0 Hz, 1H), 1.79 (s, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 195.6, 194.8, 166.2 (d, $^1J_{C-F}$ = 256.8 Hz), 151.6, 150.4, 140.3, 138.9, 136.0, 134.0, 133.2, 131.5 (d, $^3J_{C-F}$ = 9.6 Hz), 129.4, 129.2, 129.1, 128.9, 128.0, 127.9, 116.2 (d, $^2J_{C-F}$ = 22.0 Hz), 49.8, 43.5, 20.4. IR (KBr): 1661, 1596, 1240, 1153, 698 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{28}H_{22}FO_3^+$, 425.1547; found, 425.1551.

*4.4.9. (Z)-4-Benzoyl-5-methyl-6-(2-oxohexylidene)-1,6-dihydro-[1,1′-biphenyl]-3(2H)-one (**3eb**).* It is obtained as yellow oil (58.0 mg, 75%). R_f = 0.40 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 7.68 (d, J = 7.5 Hz, 2H), 7.54 (t, J = 7.0 Hz, 1H), 7.43 (t, J = 7.5 Hz, 2H), 7.38 (t, J = 7.5 Hz, 3H), 7.28 (d, J = 7.5 Hz, 2H), 6.13 (s, 1H), 4.24 (dd, J = 5, 8 Hz, 1H), 3.20 (dd, J = 8.0, 17.0 Hz, 1H), 2.98 (dd, J = 5.0, 17.0 Hz, 1H), 2.52 (t, J = 7.5 Hz, 2H), 1.85 (s, 3H), 1.60−1.54 (m, 2H), 1.34−1.26 (m, 2H), 0.89 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 202.2, 195.7, 194.9, 152.7, 150.3, 140.1, 138.6, 136.0, 134.0, 130.2, 129.1, 129.1, 128.9, 127.9, 127.8, 50.1, 43.6, 43.4, 26.0, 22.3, 20.3, 13.9. IR (KBr): 1646, 1456, 1077, 698 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{26}H_{27}O_3^+$, 387.1955; found, 387.1956.

*4.4.10. (Z)-2-(4-benzoyl-3-methyl-5-oxo-5,6-dihydro-[1,1′-biphenyl]-2(1H)-ylidene)acetaldehyde (**3gb**).* It is obtained as yellow oil (34.8 mg, 45%). R_f = 0.40 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 8.32−8.31 (m, 2H), 7.51−7.45 (m, 3H), 7.34 (t, J = 7.0 Hz, 2H), 7.19 (d, J = 7.5 Hz, 3H), 6.40 (d, J = 3.0 Hz, 1H), 5.37 (d, J = 3.0 Hz, 1H), 3.75 (dd, J = 7.5, 12.5 Hz, 1H), 3.21 (dd, J = 12.5, 29.5 Hz, 1H), 3.09 (dd, J = 7.5, 24.5 Hz, 1H), 1.88 (s, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 193.6, 158.6, 143.7, 138.0, 132.4, 128.9, 128.5, 128.4, 128.2, 127.8, 127.2, 122.7, 118.0, 106.4, 71.2, 49.8, 45.9, 19.7. IR (KBr): 1688, 1621, 1568, 1137, 698 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{22}H_{19}O_3^+$, 331.1329; found, 331.1338.

4.5. General Procedure for the Transformation of Compound 3 into Compound 4. A solution of compound **3** (0.2 mmol) in DCM (2 mL) was syringed into a stirring mixture of $BF_3{\cdot}OEt_2$ (0.2 mmol, 1.0 equiv) in a water bath at 25 °C. The reaction mixture was stirred for several hours and then quenched with saturated aqueous $NaHCO_3$. The water layer was extracted with DCM. Organic layers were combined, dried over Na_2SO_4, and concentrated in vacuum. Then, the residue was purified by column chromatography silica gel with a suitable mixture of PE and EA to afford product **4**.

*4.5.1. 1 mmol Scale for One Representative Example: Synthesis of **3aa** or **4aa** and the Transformation of **3aa** into **4aa**.* Ethyl 2-diazo-5-(2-methyl-5-phenylfuran-3-yl)-3-oxo-5-phenylpentanoate **2aa** (402.4 mg, 1 mmol) in DCM (4 mL) was syringed dropwise into a vigorous stirring solution of $Rh_2(OAc)_4$ (44.2 mg, 0.1 mmol) in anhydrous DCM (2 mL) in a water bath at 25 °C for 30 min. After the completion of the reaction (monitored by TLC), the mixture was concentrated under reduced pressure. The crude product was purified by column chromatography on silica gel with a suitable mixture of petroleum ether and ethyl acetate (ethyl acetate/petroleum ether = 1:6) to afford product **3aa** (0.75 mmol, 280.8 mg) in 75% isolated yield.

A solution of compound **3aa** (374.2 mg, 1 mmol) in DCM (5 mL) was syringed into a stirring mixture of $BF_3{\cdot}OEt_2$ (1 mmol, 1.0 equiv) in a water bath at 25 °C. After stirring for several hours, the reaction was quenched with saturated aqueous $NaHCO_3$. The water layer was extracted with DCM. Collected organic layers were dried over Na_2SO_4, concentrated in vacuum, and then purified by column chromatography on silica gel with a suitable mixture of petroleum ether and ethyl acetate to provide **4aa** (0.60 mmol, 224.5 mg) in 60% isolated yield.

*4.5.2. Ethyl 5-Hydroxy-3-methyl-2-(2-oxo-2-phenylethyl)-[1,1′-biphenyl]-4-carboxylate (**4aa**).* It is obtained as a yellow solid (57.7 mg, 78%), mp 126−128 °C. R_f = 0.50 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 10.92 (s, 1H), 7.91 (d, J = 7.5 Hz, 2H), 7.56 (t, J = 7.5 Hz, 1H), 7.43 (t, J = 7.5 Hz, 2H), 7.32−7.30 (m, 3H), 7.25−7.23 (m, 2H), 6.86 (s, 1H), 4.46 (d, J = 7 Hz, 2H), 4.25 (s, 2H), 2.41 (s, 3H), 1.42 (t, J = 7.5 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 197.9, 171.3, 159.9, 149.2, 141.2, 140.3, 136.5, 133.1, 128.5, 128.2, 128.0, 127.9, 127.5, 124.0, 116.6, 113.2, 61.6, 41.1, 19.5, 14.1. IR (KBr): 3424, 2983, 1660, 1249, 1118, 995, 868, 757, 700 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{24}H_{23}O_4^+$, 375.1591; found, 375.1597.

*4.5.3. Ethyl 5-Hydroxy-3-methyl-2-(2-oxo-2-(p-tolyl)ethyl)-[1,1′-biphenyl]-4-carboxylate (**4ba**).* It is obtained as a yellow solid (52.8 mg, 68%), mp 140−142 °C. R_f = 0.50 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 10.84 (s, 1H), 7.80 (d, J = 8.0 Hz, 2H), 7.30−7.28 (m, 3H), 7.23−7.21 (m, 4H), 6.84 (s, 1H), 4.45 (q, J = 7.0 Hz, 2H), 4.20 (s, 2H), 2.40 (d, J = 4.5 Hz, 6H), 1.42 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 197.6, 171.4, 160.0, 149.4, 144.0, 141.4, 140.4, 134.2, 129.3, 128.4, 128.3, 128.2, 127.6, 124.3, 116.8, 113.3, 61.8, 41.1, 21.7, 19.6, 14.2. IR (KBr): 3440, 2921, 1658, 1249, 1117, 807, 703 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + Na]^+$ calcd for $C_{25}H_{24}NaO_4^+$, 411.1567; found, 411.1570.

*4.5.4. Ethyl-5-hydroxy-2-(2-(4-methoxyphenyl)-2-oxoethyl)-3-methyl-[1,1′-biphenyl]-4-carboxylate (**4ca**).* It is obtained as a yellow solid (66.3 mg, 82%), mp 135−137 °C. R_f = 0.50 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 10.83 (s, 1H), 7.88 (d, J = 9.0 Hz, 2H), 7.30−7.28 (m, 3H), 7.23−7.21 (m, 2H), 6.90 (d, J = 8.5 Hz, 2H), 6.83 (s, 1H), 4.45 (q, J = 7.0 Hz, 2H), 4.17 (s, 2H), 3.86 (s, 3H), 2.39 (s, 3H), 1.42 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 196.5, 171.5, 163.5, 159.9, 149.4, 141.4, 140.5, 130.3, 129.7, 128.4, 128.3, 127.5, 124.4, 116.7, 113.7, 61.7, 55.5, 40.8, 19.6, 14.2. IR (KBr): 3443, 2923, 1658, 1601, 1255, 1171, 831, 704 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{25}H_{25}O_5^+$, 405.1697; found, 405.1697.

*4.5.5. Ethyl-2-(2-(4-bromophenyl)-2-oxoethyl)-5-hydroxy-3-methyl-[1,1′-biphenyl]-4-carboxylate (**4da**).* It is obtained as a yellow solid (61.6 mg, 68%), mp 142−144 °C. R_f = 0.50 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 10.87 (s, 1H), 7.73 (d, J = 8.5 Hz, 2H), 7.55 (d, J = 8.5 Hz, 2H), 7.30−7.29 (m, 3H), 7.20−7.18 (m, 2H), 6.83 (s, 1H), 4.46 (q, J = 7.0 Hz, 2H), 4.18 (s, 2H), 2.39 (s, 3H), 1.42 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 196.9, 171.2, 160.0, 149.1, 141.1, 140.3, 135.2, 131.8, 129.4, 128.2, 128.1, 127.5, 123.7, 116.6, 113.3, 61.7, 41.0, 19.5, 14.1. IR (KBr): 3427, 2982, 1660, 1586, 1313, 1248, 992, 810, 704 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{24}H_{22}BrO_4^+$, 453.0696; found, 453.0704.

*4.5.6. Ethyl-2-(2-(4-chlorophenyl)-2-oxoethyl)-5-hydroxy-3-methyl-[1,1′-biphenyl]-4-carboxylate (**4ea**).* It is obtained as a yellow solid (59.7 mg, 73%), mp 136−138 °C. R_f = 0.50 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 10.85 (s, 1H), 7.80 (d, J = 8.5 Hz, 2H), 7.39 (d, J = 8.5 Hz, 2H), 7.30−7.29 (m, 3H), 7.20−7.18 (m, 2H), 6.83 (s, 1H), 4.45 (q, J = 7.0 Hz, 2H), 4.18 (s, 2H), 2.39 (s, 3H), 1.42 (t, J = 7.5 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 196.8, 171.4, 160.1, 149.3, 141.3, 140.4, 139.6, 135.0, 129.4, 128.9, 128.4, 128.3, 127.7, 123.8, 116.8, 113.4, 61.8, 41.1, 19.6, 14.2. IR (KBr): 3442, 1653, 1249, 1092, 1461, 1365, 744 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{24}H_{22}ClO_4^+$, 409.1201; found, 409.1204.

*4.5.7. Ethyl-2-(2-(4-fluorophenyl)-2-oxoethyl)-5-hydroxy-3-methyl-[1,1′-biphenyl]-4-carboxylate (**4fa**).* It is obtained as yellow oil (64.4 mg, 80%). R_f = 0.50 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 10.87 (s, 1H), 7.92−7.90 (m, 2H), 7.30−7.29 (m, 3H), 7.21−7.19 (m, 2H), 7.09 (t, J = 8.5 Hz, 2H), 6.84 (s, 1H), 4.46 (q, J = 7.5 Hz, 2H), 4.19 (s, 2H), 2.40 (s, 3H), 1.42 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 196.4, 171.4, 165.8 (d, $^1J_{C-F}$ = 255.0 Hz), 160.1, 149.3, 141.3, 140.4, 133.1 (d, $^4J_{C-F}$ = 2.9 Hz), 130.4 (d, $^3J_{C-F}$ = 9.3 Hz), 128.3, 128.3, 127.7, 124.0, 116.8, 115.7 (d, $^2J_{C-F}$ = 21.9 Hz), 113.4, 61.8, 41.1, 19.6, 14.2. IR (KBr): 3441, 2983, 1663, 1599, 1240, 838, 704 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{24}H_{22}FO_4^+$, 393.1497; found, 393.1505.

*4.5.8. Ethyl 5-Hydroxy-3-methyl-2-(2-oxo-2-(thiophen-3-yl)-ethyl)-[1,1′-biphenyl]-4-carboxylate (**4ga**).* It is obtained as a yellow solid (60.9 mg, 80%), mp 130−132 °C. R_f = 0.55 (ethyl acetate/petroleum ether = 1:6). ^{1}H NMR (500 MHz, $CDCl_3$, ppm): δ 10.85 (s, 1H), 7.94 (d, J = 2.5 Hz, 1H), 7.50 (d, J = 5.0 Hz, 1H), 7.33−7.28 (m, 4H), 7.23−7.21 (m, 2H), 6.83 (s, 1H), 4.46 (q, J = 7.5 Hz, 2H), 4.14 (s, 2H), 2.42 (s, 3H), 1.42 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 192.2, 171.4, 160.0, 149.3, 141.6, 141.2, 140.5,

https://dx.doi.org/10.1021/acs.joc.9b03093
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131.8, 128.3, 128.2, 127.6, 127.0, 126.3, 123.9, 116.7, 113.3, 61.7, 42.3, 19.6, 14.1. IR (KBr): 3426, 1662, 1311, 1249, 1202, 1118, 869, 705 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{22}H_{21}O_4S^+$, 381.1155; found, 381.1164.

*4.5.9. Ethyl 5-Hydroxy-3-methyl-2-(2-oxohexyl)-[1,1′-biphenyl]-4-carboxylate (**4ha**).* It is obtained as yellow oil (59.5 mg, 86%). R_f = 0.60 (ethyl acetate/petroleum ether = 1:6). ^1H NMR (500 MHz, $CDCl_3$, ppm): δ 10.82 (s, 1H), 7.38−7.35 (m, 3H), 7.17−7.16 (m, 2H), 6.78 (s, 1H), 4.45 (q, J = 7.0 Hz, 2H), 3.64 (s, 2H), 2.38 (s, 3H), 2.27 (t, J = 7.5 Hz, 2H), 1.48−1.45 (m, 2H), 1.42 (t, J = 7.0 Hz, 3H), 1.25−1.21 (m, 2H), 0.85 (t, J = 7.5 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 208.8, 171.3, 159.8, 149.0, 141.3, 140.1, 128.3, 128.2, 127.5, 124.0, 116.6, 113.2, 61.7, 45.2, 41.9, 26.0, 22.2, 19.3, 14.1, 13.8. IR (KBr): 3423, 2960, 1717, 1461, 1247, 868, 703 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + Na]^+$ calcd for $C_{22}H_{26}NaO_4^+$, 377.1723; found, 377.1730.

*4.5.10. Ethyl-2-(3,3-dimethyl-2-oxobutyl)-5-hydroxy-3-methyl-[1,1′-biphenyl]-4-carboxylate (**4ia**).* It is obtained as yellow oil (42.5 mg, 65%). R_f = 0.60 (ethyl acetate/petroleum ether = 1:6). ^1H NMR (500 MHz, $CDCl_3$, ppm): δ 10.82 (s, 1H), 7.36−7.33 (m, 3H), 7.14−7.12 (m, 2H), 6.77 (s, 1H), 4.45 (q, J = 7.0 Hz, 2H), 3.75 (s, 2H), 2.33 (s, 3H), 1.42 (t, J = 7.0 Hz, 3H), 1.05 (s, 9H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 213.1, 171.4, 159.8, 149.1, 141.3, 140.1, 128.2, 128.1, 127.5, 124.1, 116.5, 113.2, 61.6, 44.2, 39.4, 26.8, 19.1, 14.2. IR (KBr): 3423, 2972, 1662, 1372, 1245, 867, 701 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + Na]^+$ calcd for $C_{22}H_{26}NaO_4^+$, 377.1723; found, 377.1728.

*4.5.11. 2-(4-Benzoyl-5-hydroxy-3-methyl-[1,1′-biphenyl]-2-yl)-1-phenylethanone (**4ab**).* It is obtained as yellow oil (34.1 mg, 50%). R_f = 0.30 (ethyl acetate/petroleum ether = 1:6). ^1H NMR (500 MHz, $CDCl_3$, ppm): δ 7.85 (t, J = 7.0 Hz, 4H), 7.56−7.52 (m, 2H), 7.47 (t, J = 8.0 Hz, 2H), 7.41 (t, J = 8.0 Hz, 2H), 7.31−7.30 (m, 3H), 7.26−7.24 (m, 2H), 6.85 (d, J = 1.0 Hz, 1H), 4.18 (s, 2H), 1.88 (s, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 200.4, 198.0, 155.4, 147.8, 141.2, 139.3, 138.0, 136.7, 133.3, 133.2, 129.5, 128.8, 128.6 128.6, 128.5, 128.4, 128.0, 127.6, 124.7, 123.9, 116.2, 40.6, 20.3. IR (KBr): 3423, 2974, 1679, 1451, 1242, 1365, 758, 695 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{28}H_{23}O_3^+$, 407.1642; found, 407.1650.

*4.5.12. 2-(4-Benzoyl-5-hydroxy-3-methyl-[1,1′-biphenyl]-2-yl)-1-(4-fluorophenyl)ethanone (**4cb**).* It is obtained as yellow oil (67.9 mg, 70%). R_f = 0.30 (ethyl acetate/petroleum ether = 1:6). ^1H NMR (500 MHz, $CDCl_3$, ppm): δ 7.90−7.87 (m, 2H), 7.84 (d, J = 7.0 Hz, 2H), 7.56−7.53 (m, 1H), 7.58 (t, J = 7.5 Hz, 1H), 7.48 (t, J = 8.0 Hz, 2H), 7.33−7.31 (m, 3H), 7.24−7.23 (m, 2H), 7.08 (t, J = 8.5 Hz, 2H), 6.85 (s, 1H), 4.15 (s, 2H), 1.88 (s, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 200.3, 196.3, 165.7 (d, $^1J_{C-F}$ = 255.0 Hz), 155.4, 147.6, 141.0, 139.2, 137.9, 133.3, 130.6 (d, $^3J_{C-F}$ = 9.3 Hz), 129.4, 128.8, 128.5, 128.3, 127.6, 124.4, 123.8, 116.2, 115.7 (d, $^2J_{C-F}$ = 21.8 Hz), 40.4, 20.2. IR (KBr): 3439, 1661, 1596, 1240, 1153, 839, 698 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{28}H_{22}FO_3^+$, 425.1547; found, 425.1547.

*4.5.13. 1-(4-Benzoyl-5-hydroxy-3-methyl-[1,1′-biphenyl]-2-yl)-hexan-2-one (**4eb**).* It is obtained as yellow oil (34.8 mg, 78%). R_f = 0.40 (ethyl acetate/petroleum ether = 1:6). ^1H NMR (500 MHz, $CDCl_3$, ppm): δ 8.26 (s, 1H), 7.82 (d, J = 7.5 Hz, 2H), 7.58 (t, J = 7.5 Hz, 1H), 7.48 (t, J = 7.5 Hz, 2H), 7.40−7.37 (m, 3H), 7.22−7.21 (m, 2H), 6.81 (s, 1H), 3.60 (s, 2H), 2.26 (t, J = 7.5 Hz, 2H), 1.86 (s, 3H), 1.47−1.42 (m, 2H), 1.25−1.19 (m, 2H), 0.84 (t, J = 7.0 Hz, 3H). ^{13}C {1H} NMR (125.8 MHz, $CDCl_3$, ppm): δ 208.7, 200.4, 155.4, 147.6, 141.2, 139.3, 137.7, 133.2, 129.4, 128.8, 128.5, 128.3, 127.6, 124.5, 123.6, 116.1, 44.5, 42.1, 26.0, 22.3, 20.2, 13.8. IR (KBr): 3446, 1646, 1456, 1077, 694 cm^{-1}. HRMS (ESI-TOF) m/z: $[M + H]^+$ calcd for $C_{26}H_{27}O_3^+$, 387.1955; found, 387.1960.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.9b03093.

Crystallographic information (**4ea**: CCDC 1876455) (CIF)

Optimization of the ratio of the three reactants, ORTEP drawing of 6r, crystal parameters and refinement metrics for compound 6r, and NMR spectra (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

This work was supported by the Construct Program of the Key Discipline in Hunan Province and by the Science and Technology Planning Project of Hunan Province (2018TP1017).

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https://dx.doi.org/10.1021/acs.joc.9b03093
J. Org. Chem. 2020, 85, 2395−2405